PROVISIONAL APPLICATION FOR PATENT

UNDER 35 U.S.C. § 111(b)

BEDROOM SAFETY GUARD STATE ARMED BY AN ORDINARY CUE, WITH WAKE-WORD-FREE SAFE-PHRASE LISTENING AND PRIVATE ESCALATION; ARRIVAL-TRIGGERED SAFETY CHECK-IN; ON-DEMAND REFUGE LODGING; A VERIFIABLE MEDIATED APOLOGY CHANNEL; AND PRIVACY-PRESERVING DISTANCING

Part B, version 1.0.0 — November 2, 2026.

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Related applications: U.S. Provisional Patent Application No. 64/183,657; U.S. Patent No. 12,347,301 B1; U.S. Provisional Patent Application No. 64/016,800; U.S. Patent Application Publication No. 2016/0196735 A1 (each incorporated by reference)

Entity status: Micro entity (certification on Form PTO/SB/15A filed herewith)

USPTO customer number: 208507

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is related to U.S. Provisional Patent Application No. 64/183,657, filed October 24, 2026, titled "Privacy-Preserving Ambient Consent Signaling Gated on Continuously Renewed Affirmation, with Music, Lapse Detection and Escalation to Help" (the "'657 application"), of the present inventor, the entire disclosure of which is incorporated herein by reference. Terms defined in the '657 application have the same meanings here unless this application defines them differently, and every mechanism described below may be practiced together with, or independently of, the consent-gated media controller, affirmation windows, lapse detection, escalation sequence (check-in, alarm, help), greyzone and brake signals, room tags and dispatchable bedroom locations, breath-alcohol interfaces, privacy modes, data-flow manifest, event log, evidence locker, and aftercare described there.

[0002] This application is also related to U.S. Patent No. 12,347,301 B1, titled "Consent management system over a communications network," issued to the present inventor (the "'301

patent"); to U.S. Provisional Patent Application No. 64/016,800, titled "Networked Emergency Dispatch and Preparedness Simulation System" (the "peer-response application"); and to U.S. Patent Application Publication No. 2016/0196735 A1 of the present inventor (published under the name Adam Clayman), titled "Systems and Methods for Monitoring Health in a Shared Living Environment." Each is incorporated herein by reference in its entirety. The responder alerting, location-based responder selection, responder vetting, and access and entry-management subject matter of the peer-response application may be used to provide the responder, host, and transport functions described below.

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

[0003] Not applicable.

TECHNICAL FIELD

[0004] The disclosed subject matter relates to personal-safety and consent-support systems, and more specifically to related mechanisms that work before, during, and after a risky intimate encounter. The first is a private safety check-in that a mobile device raises when its user arrives late at night at an unfamiliar residence in a context of elevated risk. The second is a system that, on a discreet request, assigns the requester a refuge for the night: a lockable safe-sleep room or a vetted volunteer host with whom there is no romantic interest. The third is a mediated channel, run by an ombudsperson or any other authority entrusted with the intermediary role, through which a person restrained by a no-contact order may submit an apology that is held for the protected person to open when he, she, or they choose, and may send structured clarifying questions that the protected person may answer or decline; with it comes a mechanism that, where requested and necessary, helps the parties to such an order avoid one another by rerouting them gently around each other without disclosing either party's location. A fourth is a guard state in which a device in the user's bedroom, armed discreetly by an ordinary cue such as a song, a routine voice request, or an ordinary device command, listens on the device without a wake word for a safe phrase and responds through a private check-in, an alarm, and a call for help.

BACKGROUND

[0005] Three moments around an intimate encounter. The '657 application, which is related to this application and is not admitted to be prior art, describes a system that is present during an intimate encounter: it keeps a shared ambient signal running only while every participant renews affirmation,

fades it upon a lapse, stops it upon a stop, and escalates an unanswered lapse through a check-in and an alarm to help. This application addresses three moments that lie outside that session and that the '657 application does not reach. The first comes before any session: the moment a person arrives, late at night and often after drinking, at an unfamiliar residence with someone. The second is the moment of exit: when a person wants to leave a partner's or host's room in the night but has nowhere safe to go. The third comes after harm has been reported: the period during which a no-contact order separates the two people and freezes every misunderstanding between them in place.

[0006] The hot-cold empathy gap. As set out in the '657 application, research on the hot-cold empathy gap (Loewenstein, 1996; Ariely and Loewenstein, 2006) shows that people in a calm, "cold" state systematically underestimate how differently they will judge and act once aroused, and that people in an aroused, "hot" state underestimate how differently they will see the same acts once calm. In the 2006 study, male participants answering while sexually aroused reported markedly greater willingness to engage in acts they had rated as unacceptable when calm, including persisting after a partner said no. A decision made in the hot state is therefore made by a person who cannot reliably reach the preferences he or she held an hour earlier. Each mechanism described here moves a decision, or an offer of help, to a moment when it can still be acted on: before the hot state, at the point of exit, or after the heat of a dispute has passed.

[0007] Freeze and fawn. Also as set out in the '657 application, the human stress response under threat takes forms commonly summarized as fight, flight, freeze, or fawn. Freeze is the involuntary stilling of body and voice (tonic immobility) reported by a substantial fraction of sexual-assault survivors; fawn is appeasement, the reflexive effort to placate a perceived threat by complying. A safety system that waits for the person at risk to recognize the danger and press a button serves the person who fights and fails the person who freezes or fawns. A system that asks, privately and early, and that makes a quiet answer as easy as a dismissal, serves all four.

[0008] The absence of a capable guardian. Routine activity theory holds that a direct-contact crime occurs when a motivated offender and a suitable target converge in the absence of a capable guardian (Cohen and Felson, 1979). A bar, a party, or a street has bystanders, staff, and lighting. A private residence has none of these, and the moment two people arrive at one together after a night out is the moment the guardian disappears. A cluster-randomized trial of the Green Dot bystander program in Kentucky high schools found reductions in sexual violence perpetration (Coker and colleagues, 2017), but bystander programs act where a bystander is present and cannot follow a person through the door. Situational crime prevention, in the form developed by Ronald Clarke and

colleagues, sorts preventive measures into five mechanisms: increasing the effort, increasing the risks, reducing the rewards, reducing provocations, and removing excuses (Cornish and Clarke, 2003). A place to go, made available before it is needed, is a means of exit of exactly this situational kind.

[0009] Alcohol. Alcohol consumption by the perpetrator, the victim, or both is present in roughly half of sexual assaults (Abbey, 2002), and the '657 application likewise describes a breath-alcohol reading taken when a participant leaves a drinking occasion and a consent lockout computed from it. Drinking can also be inferred without a breath sensor: a model trained on smartphone sensor data classified non-drinking, drinking, and heavy-drinking episodes of young adults with 96.6 percent accuracy (Bae and colleagues, 2017). A person who has been drinking is both at greater risk and less able to judge that risk, and is also at risk of harm from the alcohol itself if left to sleep alone, unobserved.

The moment of arrival

[0010] Existing personal-safety check-ins are built to confirm that a person reached a destination. In Apple's Check In feature, a user who starts a check-in for a destination is notified when he or she enters the arrival area, and the chosen partner receives a simple arrival message; if the user does not arrive in time, diverges from the route, or goes offline, the partner is alerted, and a timed check-in that goes unanswered starts a fifteen-minute countdown before the partner is alerted (TidBITS, 2023). Google's Personal Safety application offers a safety check of a chosen duration with a reason the user selects, after which the user is prompted to confirm being safe, share his or her location, or call emergency services, with one minute to choose before an alert is sent (Android Central). Kitestring, from 2014, texts the user at a planned return time and alerts emergency contacts if the user does not reply (Time, 2014). Each of these is started by the user for a trip the user has planned, and each treats arrival as the end of risk. None appears to be raised by arrival itself, and none distinguishes arrival at home from arrival, at two in the morning, at a residence the user has never visited.

[0011] Some systems infer that something may be wrong. Uber's RideCheck senses route deviations, long stops, and drop-offs short of the planned destination, and asks the rider and driver whether all is well, offering a safety toolkit or another ride (Uber, Hong Kong newsroom); its attention ends with the ride. U.S. Patent No. 10,536,570 B2 (McAfee, priority 2012) describes treating a deviation of a user's present context from the user's historical context beyond a threshold as a possible safety event; it is directed to deviations from routine generally, and does not describe a

private, dismissible check-in raised by arrival at an unfamiliar residence in a context of drinking, a night out, or a date. U.S. Patent Application Publication No. 2014/0279123 A1 (Bank of America) describes initiating a check-in when a user crosses a geofence around a merchant's location, for commerce rather than safety. A dating platform's integration with an emergency-response service, announced in 2020, lets a user record the details of a planned date and summon help with a button (The Motley Fool, 2020).

[0012] The relationship of the present mechanisms to the related '657 application, which is not admitted to be prior art, is described in section 2.

The missing exit

[0013] A person who wants to leave a partner's or a host's room in the night faces a poor choice. He or she can stay with someone who will not take no for an answer, or walk or ride home alone, often impaired, at two or three in the morning. Venue programs offer an exit only while the person is still in the venue. Under the Ask for Angela campaign, a person who asks staff for "Angela" is taken to a part of the venue out of sight of the threat and helped to leave or to get a taxi (Bradford Council). Campus safe-ride services take a student home, but home may be shared with, or near, the person being left.

[0014] Emergency hosting and refuge programs exist for other populations. Nightstop, in the United Kingdom, places young people aged 16 to 25 who are homeless or at risk of homelessness with volunteer hosts who have a spare room available at short notice; hosts are vetted by an enhanced criminal-record check, references, an interview, a home safety assessment, and a home visit, and receive daily check-in calls and a 24-hour on-call number during each placement (Nightstop host role description). Host-home programs in the United States place youth with trained, licensed volunteers for up to 21 days (Lutheran Social Service of Minnesota). The National Safe Place Network's TXT 4 HELP service replies to a youth who texts "SAFE" and a location with the address of the nearest Safe Place and a local youth agency or shelter (National Safe Place Network). Shelter-bed availability platforms let agencies find a vacant bed for the night (Finding A Bed Tonight). Residence halls increasingly use electronic locks with mobile credentials. And the federal Title IX regulations list mutual restrictions on contact among the supportive measures a school may offer after a report (34 C.F.R. § 106.30(a), as amended in 2020).

[0015] The references described above do not appear to provide a refuge for the night to a person exiting an intimate or dating encounter, requested discreetly from inside the room, assigned in

minutes without waking anyone, opened by a time-limited credential, kept unknown to the person being left, and matched, where the refuge is a volunteer's home, so that no romantic interest can arise between host and guest. Nor do they appear to account for the impaired guest who must not be left to sleep unobserved.

The frozen dispute

[0016] After a report of sexual misconduct, a school, employer, or court commonly issues a no-contact order. A campus no-contact order typically restricts verbal, nonverbal, physical, and electronic communication, including communication through third parties acting on a party's behalf (University of Utah Title IX training). The order protects the complainant. It can also freeze every misunderstanding in place for the months a proceeding takes. A respondent who wishes to apologize cannot. A respondent who believes an investigator has misread the allegation, its scope, or what outcome the complainant wants has no way to ask. A complainant cannot learn whether the respondent understands what happened. Each side's window onto the other is often an investigator's summary, and an error in that summary may go uncorrected.

[0017] Existing channels each leave a gap. Restorative-justice approaches to the informal resolution of campus sexual misconduct center the needs of the person harmed and require both parties to opt into a structured process (Orcutt and colleagues, 2020). Corrections departments run offender apology banks: the Colorado Department of Corrections, for example, screens each letter, rejects letters with re-victimizing content, retains approved letters confidentially and indefinitely, and notifies the victim that a letter is waiting, the choice to obtain it being solely the victim's; but letters submitted in cases with active no-contact protection orders cannot be approved (Colorado Department of Corrections). Co-parenting platforms used under court orders route all communication between parents through a recorded service with a tone-checking aid, and model orders provide that using the platform is not by itself a violation of a protection order (OurFamilyWizard model order language); they are built for ongoing two-way communication, not for a one-way submission held under the recipient's control. Callisto's matching system holds a survivor's entry until another survivor names the same perpetrator, then connects each to a confidential advocate (Wikipedia: Callisto). U.S. Patent Application Publication No. 2015/0339627 A1 describes structured, state-tracked communication between parties in business and legal workflows.

[0018] The references described above do not appear to provide a channel that is always open to a restrained party's submission during an active no-contact order; that holds each submission for the

protected person to open only if and when he or she chooses, including a mode in which the protected person is never notified and checks only when he or she wishes; that tells the submitter exactly what was transmitted, what an intermediary changed or withheld and why, and when it was read; that lets each party verify, independently of the intermediary, that what was received is what the other party wrote and that nothing was silently suppressed; that confines questions to closed, structured forms about the investigator's understanding; and that is run by an office independent of the investigation.

[0019] There is therefore a need for a check-in that arrives at the moment risk begins rather than when a trip ends; for a refuge that can be summoned from inside the room and reached safely; and for a channel through which an apology can always be offered and a misunderstanding corrected without contact, without pressure, and without exposing the protected person.

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SUMMARY

[0020] This Summary introduces a selection of concepts in simplified form that are further described in the Detailed Description. It is not intended to identify key or essential features of the claimed subject matter.

[0021] Arrival check-in. In one embodiment, a mobile device determines, on the device, that its user has arrived at a place; classifies the place as unfamiliar, meaning that it is not among the user's known places (home, recognized sleeping sites, and frequently visited places) and is likely a residence; determines that the arrival falls within a configured late-night window; and determines that a risk context is present, such as a breath-alcohol reading or an inferred drinking episode earlier that night, departure from a drinking occasion, a date arranged through a linked dating platform, or another person's device having traveled with the user. Only when all of these hold does the device present a private check-in: a quiet prompt on the user's own phone or wearable, dismissible in one gesture, offering graded answers (fine, a greyzone answer, and a request for help) and an enrolled duress answer that looks like an ordinary dismissal. A greyzone answer privately alerts a person the user chose in advance, offers to arm the guard state of the '657 application, and offers a way out, including a ride or a refuge for the night. Location, readings, and the classification stay on the device; nothing is shared unless the user answers in a way that asks for it.

[0022] Refuge for the night. In another embodiment, a refuge coordination service receives a discreet request for a place to sleep, made by a press pattern on a bedside or headboard button, a greyzone answer to a check-in, a phrase, or an application. It first screens the request for medical risk, so that a person who may be dangerously intoxicated is routed to medical evaluation or a monitored setting rather than to a room alone. It then assigns a refuge from an inventory that includes lockable safe-sleep rooms in residence halls and other buildings, partner facilities, and the homes of vetted volunteer hosts. A host is matched under constraints that exclude romantic interest between host and guest, including the attraction preferences each declares, any prior acquaintance or dating-platform match, and the guest's own preferences, and the match is computed without revealing either party's declarations to the other. The service issues a time-limited credential, gives wayfinding, arranges transport if needed, monitors the night without cameras, and closes the stay in the morning, while keeping the refuge's identity and location from the person the requester left.

[0023] Held-apology and clarification channel. In another embodiment, an intermediary (an ombudsperson, or any other person, office, or service entrusted with the role, preferably independent of the investigation) runs a channel between the parties to a no-contact order. The

restrained party may always submit an apology, after a cooling-off period and confidential counseling if elected. The ombudsperson screens it against published criteria, after which it is sealed for, and openable only by, the protected party, who has chosen in advance whether to be notified at once, notified once, or never notified and left to check when he, she, or they wish. The ombudsperson may withhold or change parts of a submission, with stated reasons, but the restrained party approves the final version before it is sealed, receives a transmittal record of exactly what was transmitted, what was changed or withheld, and why, and receives a read receipt, signed by the protected party's own device, when the protected party opens it. Each party's client signs what that party writes, and every submission, decision, and receipt is committed to an append-only log witnessed by parties independent of the ombudsperson and of the issuing authority. Each party can therefore verify, without relying on the intermediary's account, that the words received are the other party's own except where visibly marked as changed or withheld, and that nothing addressed to him, her, or them was silently suppressed. The restrained party may also send structured clarifying questions from a closed set, about whether the investigator's summary states the allegation correctly, what scope the protected party means, and what outcome he, she, or they want. The protected party may answer, decline, or ignore each question; an answer goes to the restrained party, and any correction the protected party adds goes where the protected party elects, including to the decision-maker. Submissions are excluded from investigators' access and, where law or policy so provides, from use as evidence, and the issuing authority may designate use of the channel as authorized communication that does not violate the order. In a related embodiment, where the protected party requests it and it is found necessary, or where both parties elect it, the parties' devices help them avoid running into each other: rotating proximity tokens that only the other party's device can recognize, protected zones checked by oblivious lookup, and private intersection of planned routes and times let each device suggest a small change of route, entrance, or timing, with decoy suggestions so that no suggestion locates the other party, a disruption budget balanced between the parties, and no location of either party sent to a server, the intermediary, or the other party. Class, work, and personal schedules feed a forecaster that recognizes recurring and one-off conflicts days or weeks ahead and resolves them with ordinary-looking accommodations, and a periodic review lets either party ask for more distance, which takes effect at once, or less, which takes effect only if both choose it, without either learning the other's choice otherwise.

[0024] A fourth mechanism is a bedroom guard state. A device in the user's bedroom, recognized from a label, a smart-home room assignment, or signals gathered during the user's sleep, is armed

discreetly by an enrolled ordinary cue, such as selecting a particular song or playlist, making a routine voice request, or giving an ordinary device command, or automatically when a second person's device is detected in the bedroom. The device carries out the ordinary response and, with no audible change, listens on the device and without a wake word for a safe phrase, including a universal emergency phrase adopted for its locale, while cancelling its own playback. A safe phrase stops the playback and starts a private check-in, then an alarm, then a call for help, each advancing only if the user does not respond. No affirmation is awaited in the guard state; where partners are early in a relationship, the affirmative consent envelope of the '657 application may apply instead, with settings the device recommends and each participant accepts. The mechanisms share a platform: the escalation sequence, help pathways, room tags, dispatchable locations, time-limited entry grants, peer responders, privacy modes, and data-flow manifest of the '657 application and the peer-response application. They connect in sequence. An arrival check-in answered with a gray-zone reply can offer a refuge. A refuge stay can close with an aftercare step. And a matter that proceeds to a no-contact order can be carried forward through the held-apology channel.

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

[0025] FIG. 1 is a block diagram of a system in which the arrival check-in, refuge coordination, and held-apology channel operate, together with the devices, services, and networks they use, according to one embodiment.

[0026] FIG. 2 is a flow diagram of the arrival check-in method, from detection of an arrival through the familiarity, time-window, and risk-context conditions to the check-in and the action taken on each answer, according to one embodiment.

[0027] FIG. 3 illustrates a check-in prompt as presented on the user's phone and wearable, with its graded answers, according to one embodiment.

[0028] FIG. 4 is a flow diagram of the refuge method, from a discreet request through medical screening, assignment, credential issuance, arrival, the overnight period, and the morning close, according to one embodiment.

[0029] FIG. 5 is a diagram of the constraints applied in matching a guest to a volunteer host, including the exclusion of romantic interest, according to one embodiment.

[0030] FIG. 6 is a schematic plan of a safe-sleep room and its fittings, according to one embodiment.

[0031] FIG. 7 is a block diagram of the held-apology and clarification channel, showing the parties' clients, the ombudsperson service, the sealed store, the witnessed log, and the access boundary that excludes investigators, according to one embodiment.

[0032] FIG. 8 is a flow diagram of a structured clarifying question, from its selection by the restrained party to the distribution of the protected party's answer, according to one embodiment.

[0033] FIG. 9 is a state diagram of a held submission, showing the transmittal record, read receipt, and closure notice returned to the submitter, according to one embodiment.

[0034] FIG. 10 is a hardware block diagram of a computing device on which a client, a service, or a part of either executes, according to one embodiment. FIG. 11 is a diagram of the signed segment commitments of a submission, showing transmitted, withheld, and replaced segments and their verification by the protected party's client against the restrained party's signature and the witnessed log, according to one embodiment. FIG. 12 is a diagram of privacy-preserving distancing between the devices of the parties to a no-contact order, including schedule-based conflict forecasting and periodic review of the degree of separation, according to one embodiment. FIG. 13 is a flowchart of the bedroom guard state, showing arming by an ordinary cue (1302), the ordinary response and entry into the guard state (1304), listening for a safe phrase without a wake word, or for a safe tap, gesture, button pattern, or other non-verbal safe input (1306), acceptance of a detection (1308), the private check-in (1310), the tiered help response (1312) with notification of a designated contact (1314), and a request to emergency services carrying a validation level (1316), according to one embodiment.

DETAILED DESCRIPTION

[0035] The following detailed description refers to the accompanying drawings. Wherever possible, the same reference numbers are used in the drawings and the following description to refer to the same or similar elements. While embodiments of the disclosure may be described, modifications, adaptations, and other implementations are possible. For example, substitutions, additions, or modifications may be made to the elements illustrated in the drawings, and the methods described herein may be modified by substituting, reordering, or adding stages to the disclosed methods. Accordingly, the following detailed description does not limit the disclosure; the proper scope of the disclosure is defined by the appended claims and their equivalents.

1. Definitions

[0036] Unless the context requires otherwise, the following terms have the meanings given here. Terms not defined here have the meanings given in the '657 application, and the following terms used in that application have, here, at least the meanings stated. "Escalation sequence" means a sequence of a private check-in to the user on the user's own device, an alarm stage in which a device sounds or displays an alarm, and a help stage in which designated help pathways (for example trusted contacts, peer responders, and emergency services) are contacted with the user's location; the sequence advances when a stage goes unanswered and halts upon the user's response. "Guard state" means a state in which a device listens, on the device and without a wake word, only for universal and enrolled safe words, retaining no audio and generating no transcript of anything else. "Greyzone" means a signal of unease or uncertainty short of a stop or an emergency, which asks for a pause or a check and accuses no one.

[0037] "User" means a person who has installed and enrolled the client application of the system on his, her, or their own device. "Requester" means a person who requests a refuge. "Guest" means a requester who is staying, or is to stay, in a refuge. "Host" means a person who offers his, her, or their home, or a room in it, as a refuge and who has been vetted as described in section 4.11. "Protected party" means a person whom a no-contact order protects; "restrained party" means a person whom a no-contact order restrains; under a mutual order each party is both. The protected party is often the complainant and the restrained party the respondent in a proceeding, but the channel of section 5 does not depend on those roles.

[0038] "Known place" means a place stored on the user's device as one the user knows, whether because the user labeled it (for example, home, work, a family member's home, or a friend's home marked as trusted), because the '657 application's sleeping-site recognition designated it, or because the user's device has recorded visits to it exceeding a configured familiarity threshold, such as at least a number of distinct visits or a cumulative dwell time over a trailing period. "Unfamiliar place" means a place that is not a known place. Familiarity may be determined at the level of a building or, where the device can resolve it, at the level of a floor, unit, or room, so that a room on another floor of the user's own residence hall may be unfamiliar although the building is known.

[0039] "Residence" means a place where people sleep, including a house, an apartment, a dormitory or residence-hall room, a fraternity, sorority, or other group house, a hotel or motel room, a short-term rental, and a room in any of these.

[0040] "Late-night window" means a configured interval of local time, for example from 10:00 p.m. to 6:00 a.m., which may differ by day of the week, by the user's own schedule, or by a policy the user adopts.

[0041] "Risk context" means one or more signals, described in section 3, indicating that the arrival follows drinking, a night out, a date, or travel in company with a person not known to the user's devices, or otherwise carries elevated risk.

[0042] "Check-in" means a private prompt presented on the user's own device and answerable in one gesture. "Graded answers" means at least an all-clear answer, a greyzone answer (also called a gray-zone or "check on me" answer), and a help answer, together with, where enrolled, a duress answer that appears to an observer to be the all-clear answer but is processed as a silent request for help, as described for the duress affirmation in the '657 application.

[0043] "Chosen contact" means a person the user designated in advance to receive a greyzone alert, such as a friend, roommate, family member, resident adviser, or peer responder of the peer-response application.

[0044] "Refuge" means a place in which a requester may sleep for the remainder of a night, apart from the place the requester is leaving, and includes a safe-sleep room, a partner facility, and a host's home. "Safe-sleep room" means a room set aside in a residence hall, group house, or other building, lockable from inside, fitted as described in section 4.6, and assigned on request. "Partner facility" means a hotel, staffed lounge, shelter, sobering center, clinic, or similar place that has agreed to accept requesters.

[0045] "No-romantic-interest constraint" means the set of matching conditions, described in section 4.5, under which a host and a guest are matched only where neither is expected to have romantic or sexual interest in the other and where they have no prior romantic or dating connection.

[0046] "No-contact order" means any order, directive, or condition, issued by a school, employer, court, agency, or other authority, that restricts communication or contact between persons, including a mutual restriction on contact and an order of protection. "Issuing authority" means the body that issued it.

[0047] "Ombudsperson" means any person, office, or service entrusted with the intermediary role described in section 5, whatever its title: for example an ombudsperson, a Title IX or human-resources office, a court clerk or court-appointed office, a victim-services office, a mediator, an advocate organization, a licensed professional, or a commercial service operating under an

agreement with the issuing authority. The ombudsperson is preferably independent of the investigation and adjudication of the underlying matter, meaning at least that it does not report to, share staff or case files with, or take direction on individual matters from the investigators or decision-makers, but independence is not required; the verifiability of section 5.11 protects the parties whether or not it holds. The ombudsperson may be assisted by automated screening but decides each matter itself.

[0048] "Submission" means an apology, a structured clarifying question, a withdrawal notice, or another item that a party submits to the channel. "Held" means stored, after screening, in sealed form until the protected party opens it, it expires, or it is destroyed under the protected party's settings. "Sealed" means encrypted so that only the protected party, or a key holder the protected party designates, can decrypt it.

[0049] "Transmittal record" means a record returned to the submitting party that states exactly what content was transmitted to the protected party, which parts of the submission were changed or withheld in screening and the criterion and reason for each, and a cryptographic commitment to the transmitted version, as described in section 5.5. "Read receipt" means a notice to the submitting party, signed with a key held on the protected party's device, that the protected party has opened an item, identifying the version opened. "Closure notice" means a notice to the submitting party that an item was closed without being opened. "Uniform acknowledgment" means, in the alternative mode described in section 5.5, a response that is identical in content, size, and timing regardless of the protected party's settings, actions, or inaction. "Intermediary" means the ombudsperson, the issuing authority, and any operator of the service that carries the channel.

2. System overview (FIG. 1)

[0050] Referring to FIG. 1, a system 100 includes a user device 102, such as a smartphone, and optionally a wearable 104, such as a watch or ring, each running a client application 110 of the system; a network 106; a refuge coordination service 120; a peer-response network 130 of the kind described in the peer-response application; and an ombudsperson service 140. The client application 110 on the user device 102 contains an arrival check-in module 112, a place model 114, and a risk-context engine 116, each running on the device. A breath-alcohol sensor 170 and a bedside or headboard button 180, of the kinds described in the '657 application, and a room hub 182 may communicate with the user device 102 over a local link 108.

[0051] The refuge coordination service 120 maintains a refuge inventory 122 of safe-sleep rooms and partner facilities, a host registry 124 of vetted hosts, and a matching engine 126. It communicates with an access-control system 128 that commands electronic locks and issues time-limited credentials, with a transport interface 129 to ride services, and with a responder console 132 used by on-call staff, such as a resident adviser or a duty officer.

[0052] The ombudsperson service 140 maintains a case registry 142, a screening workspace 144, and a sealed store 146. It serves a client 148 of the protected party and a client 150 of the restrained party, and it is separated by an access boundary 152 from any investigator or decision-maker system 154, which has no read access to the screening workspace 144 or the sealed store 146.

[0053] Chosen contacts 160 receive greyzone alerts. Each component may be implemented on the computing device of FIG. 10. Each may be operated by a school, employer, municipality, venue, nonprofit, or commercial provider, alone or together, and the services 120 and 140 may be operated by different entities, as the independence of the ombudsperson requires.

[0054] The three mechanisms are described in turn in sections 3, 4, and 5. They may be practiced separately, together, or with any of the mechanisms of the '657 application. Every data flow between components is declared in the data-flow manifest described in the '657 application, and the privacy modes of that application (for example, Extreme Privacy and Balanced Privacy) apply to the client application 110. The '657 application, which is not admitted to be prior art, describes a consent session and the escalation sequence within it, the recognition of a user's own bedroom and regular sleeping sites, a breath prompt on departure from a drinking occasion, and an optional date timer; the arrival check-in of section 3 is raised instead by arrival at a residence that is not among the user's known places, the refuge of section 4 supplies a place to go, and the channel of section 5 operates after a matter has led to a no-contact order.

3. Arrival check-in (FIGS. 2 and 3)

[0055] Purpose. The arrival check-in asks one quiet question at the moment risk begins: when the user has just arrived, late at night, at a residence he, she, or they do not know, after drinking, a date, or a night out. It is built so that answering is as easy as dismissing, so that a person who would never press a panic button can still say, privately, "check on me." It is opt-in, set up in a calm moment, and runs entirely on the user's device.

[0056] Overview of the method. Referring to FIG. 2, a method 200 comprises: detecting an arrival 202; classifying the place as known or unfamiliar 204; classifying the place as a residence or not

206; testing whether the arrival falls within the late-night window 208; computing a risk-context score 210; applying suppression rules 212; presenting the check-in 214; and acting on the answer, which may be all-clear 216, greyzone 218, help 220, duress 222, or no answer 224. Steps 204 through 212 may run in any order, and any step whose result makes a check-in impossible ends the method early, so that most arrivals are discarded after the cheapest test.

3.1 Detecting an arrival

[0057] The arrival check-in module 112 detects an arrival 202 when the user device 102 passes from travel to a stationary state at a place and remains there for a dwell period, for example three to ten minutes. The module uses the operating system's visit-detection, significant-location-change, or activity-recognition services where available, which report arrivals at low power, and otherwise fuses satellite positioning, Wi-Fi association, cellular changes, and the inertial measurement unit's classification of walking, riding, and stillness. An arrival is also detected when a ride booked through a ride service ends, when the user's device joins a Wi-Fi network it has not joined before, or when the barometric pressure sensor indicates a climb of one or more floors after a ride or walk, consistent with entering a multi-story residence.

[0058] The module estimates the place to the precision available: a building from satellite and Wi-Fi positioning, a floor from barometric pressure calibrated as described in the '657 application, and a room or unit where a room tag or beacon of the '657 application is present. The estimate is held in memory on the device and is not written to a location history.

3.2 Classifying the place as known or unfamiliar

[0059] The place model 114 is a store, on the device and encrypted at rest, of the user's known places. Each entry holds a coarse geometry (for example, a geohash cell or a building footprint), optionally a floor or room designation, a label where the user gave one, and familiarity statistics such as the number of distinct days on which the device dwelt there and the cumulative dwell time over a trailing period, for example 90 days. The model is seeded from the user's labeled places, from sleeping sites recognized under the '657 application, and from the user's own home, work, and school locations as the user confirms them.

[0060] At step 204 the module looks up the arrival place. The place is known if it matches an entry whose familiarity exceeds a threshold, for example visits on at least three distinct days or a cumulative dwell of at least ten hours, or which the user labeled as trusted. Otherwise it is unfamiliar. Where the device resolves the floor or room, familiarity may be judged at that level: a

user who arrives at 1:30 a.m. at her own residence hall but on a floor she has never visited, as indicated by the barometric estimate, is at an unfamiliar place for the purpose of the check-in. A place visited once before is unfamiliar under the default threshold, so that a second visit to a new partner's home is still treated with care.

[0061] The user may mark any place as trusted, for example a sibling's home or a partner of long standing, from the check-in itself or from settings, and the user may mark a known place as one at which check-ins should continue, for example a group house where parties are held.

3.3 Classifying the place as a residence

[0062] At step 206 the module estimates whether the place is a residence, using signals available on the device without disclosing the user's position to a server: map data cached on the device, including building footprints and land-use or point-of-interest categories, from which a place with no commercial point of interest in a residential zone is likely a residence; the names and types of nearby Wi-Fi networks, residential routers being distinguishable from enterprise and venue networks by their naming patterns, number, and security settings; the time of day and the dwell itself, a long dwell after midnight being characteristic of a residence; and the floor estimate, a climb to an upper floor being characteristic of apartments, dormitories, and hotels. Where a server lookup is used, it is made in Balanced Privacy mode only, declared in the manifest, and reduced to a coarse cell that does not identify the building. Hotels, motels, short-term rentals, and dormitory rooms count as residences. Where the classifier is uncertain, the place is treated as a residence, because a superfluous quiet prompt costs little.

3.4 The late-night window

[0063] At step 208 the module tests whether the arrival time falls within the late-night window. The default window may be set by the user, by a preset (for example, 10:00 p.m. to 6:00 a.m. on Friday and Saturday nights and midnight to 6:00 a.m. on other nights), or by a policy the user adopts from a school or program. The window may widen during periods of known elevated risk that the user or an institution designates, such as the first weeks of a school term or a festival weekend.

3.5 The risk context

[0064] At step 210 the risk-context engine 116 computes a score from signals of the preceding hours, each weighted, each processed on the device, and each optional. The signals include the following.

- (a) A breath-alcohol reading from the breath-alcohol sensor 170 at or above a configured level within a configured period, for example 0.04 g/dL within the preceding six hours, including the leaving-the-party reading of the '657 application.
- (b) A drinking episode inferred by an on-device model from the phone's and wearable's sensors, such as movement, gait, and the time and place pattern of the evening, of the kind described by Bae and colleagues (2017).
- (c) Departure from a drinking occasion, detected as in the '657 application by the user's check-out, a venue's exit station, geofence departure from a bar, club, or registered party, or a scheduled prompt.
- (d) A date: a match token and meeting handshake of a linked dating platform under section 8D of the '657 application earlier that evening, or a calendar entry the user marked as a date.
- (e) Co-travel: another device having remained near the user's device from the place of departure to the place of arrival, as indicated by a meeting handshake of the '657 application, by an exchange of ephemeral tokens between two devices running the system, or by the persistence over the trip of short-range wireless signals of a single nearby device, judged within short windows without identifying that device or retaining its identifiers; devices of the user's own and devices of the user's chosen contacts are excluded.
- (f) A ride that ended at the arrival place rather than at the user's home, or a shared ride on which another rider's stop was skipped.
- (g) A night-out mode the user turned on earlier, or a request from a chosen contact, made as described for the touch message of the '657 application, that the user's devices offer a check-in.
- (h) An impairment flag raised under section 6A of the '657 application.

[0065] The engine combines the signals into a score, for example a weighted sum with weights set by a preset and refined as described in section 3.10, and compares it with a threshold. A user may instead elect a simple rule, such as "check in whenever I arrive somewhere new late at night," which removes the risk-context test entirely. No signal is ever sent off the device for scoring.

3.6 Suppression rules

[0066] At step 212 the module suppresses the check-in where the user has already answered one at the same place that night, where the user is on a call or driving, or where the user snoozed check-ins for the night in advance. Suppression never applies after a greyzone answer that night; once the user

has said "check on me," later check-ins are offered at the configured follow-up interval until the user gives an all-clear or the night ends.

3.7 Presenting the check-in (FIG. 3)

[0067] Referring to FIG. 3, the check-in 300 is presented at step 214 as a notification on the user device 102 and, where present, as a haptic tap on the wearable 104. Its visible text 302 is neutral, for example "Everything good?", or, at the user's election, disguised as an ordinary notification of the user's choosing, so that a person glancing at the screen sees nothing that suggests a safety tool. It makes no sound unless the user so elects. It offers controls for the all-clear answer 304, the greyzone answer 306, and the help answer 308, and a control 310 that opens further options. On the wearable the answers may be given by a tap pattern or a crown gesture enrolled in advance 314, each confirmed by a distinct haptic pulse 312, as described for button patterns in the '657 application. A press pattern on the bedside or headboard button 180 may also answer a check-in.

[0068] The check-in may be presented immediately after the dwell period, or after a delay the user sets, and the user may set a second check-in a configured interval later, for example 45 minutes, so that a situation that turns after arrival is also caught.

3.8 Acting on the answer

[0069] All-clear 216. The check-in closes. The user may, in the same gesture, mark the place as trusted, or snooze check-ins at that place for the night. Nothing is sent anywhere.

[0070] Greyzone 218. The module privately alerts the chosen contacts 160, in words the user approved in advance (for example, "Sam asked you to check on them"), together with whatever the user elected to share: nothing more, a coarse area, the place's name, a precise location, or a live location for a limited time. It then offers the user, on the user's own device, a short set of options: arming the guard state or the bedroom mode of the '657 application, so that the user's device listens on the device for universal and enrolled safe words; a pretext call or message, timed by the user, that gives a reason to leave; a ride, prefilled with the user's home or a refuge as destination; a request for a refuge under section 4; and a walking route with live sharing to a chosen contact. It schedules a follow-up check-in. The chosen contact receives options to call or message the user, to ask a peer responder of the peer-response network 130 to check in person, or to request a location share, which the user must approve unless the user elected to share it in advance.

[0071] Help 220. The module invokes the help stage of the escalation sequence of the '657 application, with the dispatchable location estimated at step 202, including the floor, so that a

trusted contact, a peer responder, or emergency services can find the user. Where the user designated an authorized responder and the place's access devices permit it, the time-limited entry grant of the '657 application may apply.

[0072] Duress 222. Where the user has enrolled a duress answer, for example a long press of the all-clear control, a particular emoji reply, or an all-clear given with a second tap, the visible behavior is identical to the all-clear answer, and the module silently invokes the duress help pathways of the '657 application, such as a trusted contact, a domestic-violence advocate or hotline, or a peer responder instructed to arrive on an ordinary pretext.

[0073] No answer 224. Because a user who has gone to sleep or put the phone away will often not answer, no answer does not by default raise an alarm. The user chooses in advance among: doing nothing; repeating the check-in once after a configured interval; telling the chosen contact that a check-in went unanswered, with or without location as elected; or, for a user who wants the strongest setting, advancing through the alarm and help stages of the '657 application.

3.9 Discretion in the room

[0074] Nothing about the check-in is announced in the room. No speaker in the room is used unless the user answers with help and has elected an audible alarm. The prompt, the answers, and the alerts appear only on the user's own devices and on those of the chosen contacts. A second person present cannot learn from the user's devices that a check-in occurred unless the user shows them.

3.10 Learning, privacy, and institutional settings

[0075] The module may learn, on the device, from the user's answers: a place answered all-clear repeatedly may be proposed as trusted, and weights in the risk-context score may be adjusted toward the user's own pattern. The module never learns to suppress check-ins from a greyzone, duress, or help answer.

[0076] The place model, the risk signals, the score, and the answers stay on the device. Where the user elects to contribute to a community count, the client contributes a count of check-ins and of greyzone answers per reporting interval through the community incidence reporting described in section 7A of the '657 application, using local differential privacy or secure aggregation, without time, place, or identity.

[0077] A school, employer, or program may publish a default configuration for its members, for example check-ins enabled for first-year students during the first weeks of term, which a member may adopt or decline. The institution receives no individual check-ins, answers, or locations.

3.11 Variations

[0078] The check-in may be raised for arrival in a vehicle parked at an unfamiliar residence, for arrival at a hotel, or upon the end of a ride at an address other than the destination entered. In a host-side variation, the check-in is raised for a user at his, her, or their own home when the device detects, late at night and in a risk context, that an unfamiliar device has arrived with the user, so that the resident, too, is asked privately whether all is well and offered the guard state. In a further variation, a dating platform linked under section 8D of the '657 application supplies the risk context for a date without receiving any location or answer.

4. Refuge for the night (FIGS. 4 to 6)

[0079] Purpose. When a partner will not take no for an answer, or a night simply turns wrong, a person may otherwise face a choice between staying and going home impaired at two in the morning. The refuge coordination service 120 gives that person somewhere safe to sleep, summoned quietly from inside the room, reached safely, and unknown to the person left behind, so that both people separate and rest.

[0080] Overview of the method. Referring to FIG. 4, a method 400 comprises: receiving a request 402; screening for immediate danger 404 and, if present, routing to the help stage 424; screening for medical risk 406 and, if present, routing to medical evaluation or a monitored refuge 426; identifying candidate refuges 408; selecting a refuge and, for a host, matching under the constraints of FIG. 5, 410; confirming with the requester 412; issuing a credential and wayfinding 414; arranging transport where needed 416; verifying arrival 418; supervising the overnight period 420; and closing the stay in the morning 422.

4.1 Making a request

[0081] A request 402 may be made by any of several channels, each chosen to be usable discreetly in the presence of another person: a press pattern on the bedside or headboard button 180 enrolled as "I need a place to sleep" or bound to the greyzone signal; the refuge option offered after a greyzone answer to an arrival check-in under section 3.8; a control in the client application 110; an enrolled phrase recognized on the device, which may be an innocuous sentence; a text message to a short code, usable without the application, in the manner of a text-for-help service; a call to a staffed line; or a request entered on the responder console 132 by a resident adviser, peer responder, or venue staff member acting at the person's request, for example after a person has asked venue staff for help under a coded-request program.

[0082] The request carries as little as the service needs: the requester's enrolled, pseudonymous identity; the dispatchable location of the requester, including the floor where known, as described in the '657 application; the number of people (the requester alone, or with a friend); whether transport is needed; accessibility needs; and answers to two short questions, "Are you in danger right now?" and "Do you feel very drunk or unwell?", each answerable with one tap. A breath-alcohol reading taken in the preceding hours may be attached at the requester's election. The requester may add preferences, described in section 4.5. Where the request is made by a button press alone, the service sends the two questions to the requester's device and proceeds on the press alone if they go unanswered, treating the requester as possibly impaired.

4.2 Screening for danger and for medical risk

[0083] Danger 404. Where the requester answers that he, she, or they are in danger, or where the request is made by a press pattern bound to help, the service first invokes the help stage of the '657 application, so that a responder comes to the door, and arranges the refuge after the requester is safe.

[0084] Medical risk 406. A person who may be dangerously intoxicated is preferably not sent to sleep alone. The service applies a rule set, published by the operator and reviewed by a clinician, that routes the request to medical evaluation, to a sobering center, or to a monitored refuge 426 (a room with an awake, trained attendant) when any of the following is present: a breath-alcohol reading at or above a configured level, for example 0.15 g/dL, or the level the operator's clinician sets; an answer indicating vomiting, inability to stand, or confusion; failure to answer the screening questions after a request made by a person who could answer them moments earlier; or a suspicion of a drugged drink, which also prompts an offer of rapid testing and of medical care. The requester is told, in the words of any medical-amnesty policy that applies, that seeking help will not be used against him, her, or them.

4.3 The refuge inventory

[0085] The refuge inventory 122 lists safe-sleep rooms and partner facilities, each with its location, capacity, accessibility, the access devices on the path to it, and a state: available, reserved, occupied, awaiting cleaning, or out of service. The state of a safe-sleep room is reported by its electronic lock, by its non-imaging presence sensor, and by housekeeping staff, so that the inventory is current without anyone entering the room. Partner facilities report availability through an interface, or hold a pre-agreed number of rooms that the service may assign directly, paid by pre-arranged vouchers.

[0086] The host registry 124 lists vetted hosts, each with the location of the host's home, the guest room's capacity and accessibility, the household's composition, pets, languages, and the host's availability, which the host declares as recurring windows (for example, Friday and Saturday nights) and as an on-call toggle. A host who is on call receives requests; a host who is not receives none.

4.4 Selecting a refuge

[0087] At step 408 the service identifies candidate refuges reachable from the requester's location within a configured time, preferring, in a default order, a safe-sleep room within walking distance by a lit and populated route; a partner facility; and a host's home. A requester may change the order, for example by asking for a room rather than a person's home.

[0088] At step 410 the service ranks the candidates by travel time and safety of the route; by the requester's preferences; by separation from the place being left, excluding a refuge in the same building or on the same floor as the requester's current location where the requester is leaving that building, and, where the requester names it, any building associated with the person being left; by accessibility and capacity; and, for hosts, by an even distribution of requests among hosts. A host candidate must also satisfy the constraints of section 4.5.

4.5 Matching a guest to a host (FIG. 5)

[0089] Referring to FIG. 5, the matching engine 126 applies constraints 500 to each pair of a requester and a host. The no-romantic-interest constraint is the core of them.

[0090] Declared attraction 502. At enrollment, each host declares, privately to the service, the genders (or other categories the operator offers) of persons toward whom the host may feel romantic or sexual attraction, and the host's own gender. Each requester may make the same declarations, or may decline to. A pair is excluded if the host may be attracted to persons of the requester's gender, or, where the requester has declared, if the requester may be attracted to persons of the host's gender. Where the requester has not declared, the requester may instead state the genders of host he, she, or they will accept.

[0091] Prior connection 504. A pair is excluded if the host and the requester have a prior romantic or dating connection, or any prior connection the requester wishes to avoid. The service determines this without either side's contacts or match history leaving its custody in readable form: each party's client, or a linked dating platform, contributes a set of keyed hashes of pseudonymous identifiers of persons with whom that party has matched or corresponded, together with a keyed hash of that

party's own pseudonymous identifier, and the service tests, by private set intersection, whether either party's own keyed hash appears in the set contributed for the other, learning only whether it does. A host is likewise excluded if he, she, or they have hosted the requester before and either reported a concern.

[0092] Household type 506. Where the constraint of paragraph [0090] cannot be satisfied, or as an additional assurance, a requester may accept, or require, a household of a kind the operator defines as lower-risk, for example a household of two or more adults who are present overnight, or a host who is a trained peer responder of the peer-response network 130.

[0093] Guest preferences 508 and logistics 510. The requester's stated preferences (for example, a host of a stated gender, no pets, a quiet home, a shared language) and the logistics (distance, capacity, accessibility) are applied as further constraints or ranking terms.

[0094] Safety status 512. A host is eligible only while the host's vetting, training, and inspection are current, while the host has no open concern under review, and while the host's guest room meets the fittings of section 4.6.

[0095] Separation 514. A host whose home is the place being left, or whose household includes the person being left, is excluded; where the requester names a person to avoid, a host connected to that person by a prior connection test of the kind in paragraph [0091] is excluded.

[0096] Private evaluation 520. The declarations of paragraph [0090] and the hashed sets of paragraph [0091] are stored encrypted and evaluated within a trusted execution environment, or by a secure computation, so that the service operator's staff cannot read them in the ordinary course and neither party learns any declaration of the other beyond what is implied by the match itself. Each party learns only that a match satisfies the constraints. The output is a ranked candidate list 530.

4.6 The safe-sleep room (FIG. 6)

[0097] Referring to FIG. 6, a safe-sleep room 600 has a door 602 with an electronic lock that cannot be slipped or twisted open from outside and that accepts time-limited credentials from the access-control system 128; an inside release 604 that unlocks and opens the door in one motion, for fire egress; a bed 606; a help button 608 at the center of the headboard, in the standardized location described in the '657 application, which alerts the responder console 132 with the room's location; a room tag 610, as described in the '657 application, that gives the room a dispatchable location; a non-imaging presence and respiration sensor 612, such as a millimeter-wave radar, that reports occupancy and, where the guest consents or the medical screen so requires, breathing; a room hub

614 that connects the button, the sensor, and, where the lock does not report itself, the lock's state to the service over Wi-Fi or cellular with battery backup; and basic supplies 620, such as water, a phone charger, a blanket, a hygiene kit, and a printed card of resources. The room has no camera and no microphone. Staff entry uses a logged override, reviewed afterward. Where a host's home is used, the guest room is fitted to the same standard as nearly as the home allows, and at least with an inside lock, a help button, and a room tag, and without any camera.

[0098] The room is located where a requester can reach it without waking roommates, passing the door of the place being left, or being seen by many people, for example on a separate corridor with its own stair access.

4.7 Credential, wayfinding, and transport

[0099] At step 412 the service shows the requester the assigned refuge in neutral terms and asks for confirmation; the requester may decline it and ask for another. At step 414 the access-control system 128 issues a credential bound to the requester's device and to the refuge's path, which may include an exterior door, an elevator floor, and the room door, each granted in turn as described for entry grants along a path in the '657 application. The credential is valid from issuance until a check-out time the next morning, and it is revoked at check-out or on the requester's request. The client shows a route that avoids, where possible, the place being left, and may share the requester's progress with a chosen contact until arrival.

[0100] Where the refuge is not within walking distance, at step 416 the transport interface 129 books a ride, paid by the operator, for the requester alone. The ride's destination is entered by the service, so that it is not shown on the requester's lock screen. The trip is watched for the anomalies of a route deviation or a long stop, and a deviation or a long stop prompts a check-in.

4.8 Arrival and the overnight period

[0101] At step 418 arrival is verified when the requester's credential opens the room, or, at a host's home, when the requester's device and the host's device complete a short-range handshake or the host scans a code the requester's device displays, in the manner of the meeting handshake of the '657 application. The host learns the guest's first name and expected arrival time; the guest learns the host's first name, household type, and address after confirming.

[0102] During the overnight period 420 the guest may lock the door and sleep. Where the medical screen indicated possible impairment below the threshold for medical routing, the service schedules welfare checks: a knock by on-call staff at configured intervals, or, with the guest's consent,

monitoring of respiration by the sensor 612, with an alert to the responder console 132 if breathing is absent or abnormal for a configured period, as described for respiration sensing in the '657 application. A press of the help button 608, or of the guest's own help controls, reaches the responder console at once. A host may likewise summon a responder at any time.

4.9 The morning close

[0103] At step 422, at a configured check-out time or when the guest leaves, the credential is revoked and the room set to awaiting cleaning. The guest receives one private message asking how he, she, or they are, and an aftercare step of the kind described in the '657 application, presenting once, neutrally, the options available (medical care, including time-limited options such as emergency contraception, post-exposure prophylaxis, and a forensic examination; counseling; reporting channels; or nothing), and offering a ride home. A host may send the operator a brief, safety-only note; guests are never rated.

4.10 The person left behind, and the record

[0104] The service never tells anyone where a requester went. It does not notify the person left behind, unless the requester chooses to send a neutral message, which the client offers to compose (for example, "I went home. Let's talk tomorrow."). The responder console shows on-call staff only what they need to act.

[0105] The service keeps the minimum: the stay's dates, the refuge type, and whether any help was summoned, under a retention period the operator publishes. A stay is not used to discipline the requester for drinking or for any conduct disclosed in the course of it, and it is disclosed to an investigator only at the requester's election. Counts of stays may contribute to the privacy-preserving community counts of the '657 application.

4.11 Capacity, misuse, and host care

[0106] The service may forecast demand from the calendar (term start, holidays, major events) and ask hosts to go on call accordingly. A request is never refused while a requester reports danger; otherwise, to prevent use as a free hotel, an operator may set a soft limit on stays per term, after which a request is reviewed by staff rather than refused. Hosts are vetted (for example, by a criminal-record check, references, an interview, and an inspection of the home), trained in trauma-informed hosting, boundaries, the signs of alcohol poisoning, the recovery position, and when to call emergency services, and re-vetted periodically. A host may receive a modest stipend or other incentive within the limits that preserve any volunteer liability protection the law provides.

5. Held-apology and clarification channel (FIGS. 7 to 9)

[0107] Purpose. A no-contact order protects the protected party, and the channel is designed not to weaken that protection. The channel exists because the order can also freeze every misunderstanding in place: the restrained party cannot apologize or ask what is alleged, and the protected party cannot learn whether the restrained party understands. The channel lets an apology always be offered and a misunderstanding corrected, while the protected party alone decides whether, when, and how anything reaches him, her, or them, and while nothing about the protected party leaks back to the restrained party.

5.1 Parties and architecture (FIG. 7)

[0108] Referring to FIG. 7, the channel 700 comprises the client 150 of the restrained party, the client 148 of the protected party, and the ombudsperson service 140, which holds the case registry 142, the screening workspace 144, and the sealed store 146. An automated pre-screen 702 assists the ombudsperson. A key holder 704 holds the key that opens sealed submissions: by default the protected party's own device, and alternatively, at the protected party's election, an advocate the protected party designates or a hardware security module that releases the key only to the protected party's authenticated request. A counselor 706, outside any reporting line to investigators, may assist the restrained party. An append-only audit log 708 records events without content, and a witnessed log 712, countersigned by witnesses 714 independent of the ombudsperson and the issuing authority, records commitments to submissions, decisions, and receipts, as described in section 5.11. A decision-maker inbox 710 receives only what the protected party elects to send to it. The access boundary 152 separates all of these, other than the decision-maker inbox 710, which lies on the decision-maker's side of the boundary and receives only what the protected party elects, from the investigator and decision-maker system 154, which cannot read the screening workspace, the sealed store, or the audit log's entries about individual submissions.

[0109] When the issuing authority issues a no-contact order, it creates an entry in the case registry 142 identifying the order, the parties by pseudonymous identifiers, the issuing authority, the order's terms and expiration, and whether the order is mutual. The order itself, or the issuing authority's policy, may designate use of the channel as authorized communication that does not violate the order, while every other contact remains prohibited. The registry records any such designation, and each client displays it to its party. Under a mutual order the channel is symmetric: each party may submit, and each party's settings protect him, her, or them.

5.2 The protected party's settings

[0110] At issuance, and at any time afterward, the protected party chooses a notification mode: (a) notify me of each new item; (b) notify me once that items are waiting, and not again; (c) never notify me, and I will check when I wish; or (d) notify my advocate, who will decide when and how to tell me. The default, where the protected party makes no choice, is (c). Neither using nor declining the channel is a condition of, or a credit in, any proceeding, measure, or outcome, except as the protected party elects under [0122], and a protected party's choice not to receive, open, or answer anything is never treated as a factor against that party. The protected party also chooses whether to receive apologies, clarifying questions, or both; an expiration for held items, or none; and whether held items are to be destroyed unopened. These settings govern delivery only. They never prevent the restrained party from submitting. The settings themselves are not disclosed to the restrained party, although read receipts and closure notices (section 5.5) reveal whether and when an item was opened; the protected party is told so before choosing settings and before opening any item.

5.3 Composing and submitting an apology

[0111] The restrained party composes an apology in the client 150. Before writing, the restrained party may elect a confidential session with the counselor 706, and the client offers published guidance on what an apology in this channel may contain: acknowledgment of the harm as the writer understands it, responsibility without justification, and nothing that asks anything of the reader. The client tells the restrained party, before submission, the evidentiary status of submissions under the rule that applies (section 5.8).

[0112] A submitted apology first enters a cooling-off period 904, for example 48 hours, during which the restrained party may revise or withdraw it. The cooling-off period moves the decision out of a hot state, in the sense of the hot-cold empathy gap, into a cold one. The operator may limit frequency, for example to one apology in a 30-day period, in which case a later apology replaces an earlier one that has not yet been opened, and the replacement is itself noted only in the audit log. Text is the default form; audio or video may be allowed by policy, subject to the same screening.

5.4 Screening

[0113] After the cooling-off period, the submission passes to the screening workspace 144. The automated pre-screen 702, which may be a rules engine, a classifier, or a language model running on the ombudsperson's own infrastructure, flags content against published criteria: threats or

intimidation; requests of any kind, including requests to withdraw a report, change testimony, or meet; pressure or inducements, such as offers of money or gifts; blame-shifting or disparagement; sexual content; information about the protected party's whereabouts or activities; messages from or for third parties; and excessive length. The ombudsperson reviews every submission and decides: accept unchanged; accept with changes, withholding particular passages or, where policy permits, replacing them, with the criterion and reason stated for each; return for revision, with the criteria not met stated to the restrained party; or reject in full, with the reasons and the passages on which the rejection rests stated to the restrained party, and, where the content is threatening or retaliatory, a referral to the issuing authority under its rules on retaliation. Under accept with changes, the restrained party is shown a comparison of the submission as written and the version proposed for transmission, each withheld or replaced passage marked with its reason, and the version is sealed only upon the restrained party's signed approval; the restrained party may instead revise or withdraw. The ombudsperson never adds words to the restrained party's text except as passages visibly marked as the ombudsperson's own.

[0114] On acceptance, and after any approval the changes require, the service encrypts the approved version, together with the signatures and commitments described in section 5.11, to the key held by the key holder 704, records a cryptographic hash of the sealed submission and the decision in the audit log 708 and in the witnessed log 712, and destroys the plaintext in the screening workspace. From then on, no one at the ombudsperson service can read the submission. The protected party can later verify, from the hash, that what he, she, or they open is what was screened.

5.5 Transmittal record and read receipts

[0115] The restrained party is entitled to know what became of what he, she, or they wrote. Upon sealing, the service returns a transmittal record 930 containing the exact text transmitted; a comparison against the text as written, showing each passage withheld or replaced, the criterion applied, and the reason; the time of sealing; and the commitment of the transmitted version in the witnessed log 712. Where a submission is rejected in full, the restrained party receives the reasons and the passages on which the rejection rests. When the protected party opens an item, the protected party's client returns a read receipt 932, signed with a key held on the protected party's device, stating the time of opening and the commitment of the version decrypted, so that the restrained party learns from the protected party's own device, rather than from the intermediary, what was read. When an item expires or is destroyed unopened, the restrained party receives a closure notice 934 stating only that the item was closed without being opened. Receipts and notices may be

released in periodic batches, for example daily, so that their timing does not track the protected party's activity minute by minute.

[0116] Beyond receipts and notices, the restrained party learns more only if the protected party chooses to tell: by answering a clarifying question (section 5.7), or by sending one of a small set of structured responses to an apology, such as "received," "accepted," "not accepted," or "please do not send more," each optional and signed by the protected party's client. In an alternative mode, read receipts and closure notices are withheld and the restrained party receives instead a uniform acknowledgment, identical in content and size for every submission in the same screening state and released at fixed times, so that nothing about the protected party's settings or actions can be inferred; the transmittal record of what was transmitted is provided in either mode.

[0117] A restrained party may send a withdrawal notice for an apology. The apology is not deleted: if unopened, it is marked "withdrawn by the sender" and the protected party sees the mark if he, she, or they ever open it; if already opened, the protected party sees the notice. The withdrawal notice is listed in the transmittal record, and the read receipt shows whether the protected party opened the apology before or after the notice.

5.6 Opening

[0118] When the protected party chooses to look, the client 148 shows what is waiting by kind (apology or question) and date, and the protected party may open any item, alone or with an advocate. The client fetches the sealed item, the key holder 704 releases the key to the protected party's authenticated request, and the item is decrypted on the protected party's device. Before the first item is opened, the client tells the protected party that opening an item sends a read receipt to the restrained party. Before displaying an item, the client verifies the restrained party's signature and the item's commitments in the witnessed log, as described in section 5.11, and displays each withheld or replaced passage as marked, with its reason, so that the protected party sees where and how the intermediary intervened, though not the withheld words. The protected party may keep, export, or destroy the item.

5.7 Structured clarifying questions (FIG. 8)

[0119] Referring to FIG. 8, a method 800 lets the restrained party ask, and the protected party answer, a small set of questions that correct misunderstandings without opening a conversation. At step 802 the restrained party selects a question from a closed set of templates; there is no free-text question. At step 804 the restrained party fills any restricted slot the template allows, such as

selecting an item from the list of charges in the notice the restrained party received. The templates include:

- (Q1) Summary accuracy: "The investigator's summary of the allegation, as provided to both parties, reads as follows: [summary]. Is this summary accurate?" Answers: accurate; partly accurate; not accurate; decline to answer.
- (Q2) Scope: "Which of the following are part of what you are raising?" with the charge items as a checklist, and "decline to answer."
- (Q3) Outcome: "What outcome would you want?" answered on the graduated scale and range of response described in the '657 application, from no formal action or a repair conversation to a maximum, or "decline to answer."
- (Q4) Future dialogue, if the operator offers it and the protected party has not disabled it: "Would you ever be open to a facilitated restorative conversation?" Answers: yes; maybe later; no; decline to answer.

[0120] At step 806 the service applies limits, for example no more than three questions in a 30-day period and no repetition of a question already answered or declined. At step 808 the ombudsperson reviews the selection for pattern or pressure. At step 810 the question is sealed and held exactly as an apology is, subject to the protected party's settings.

[0121] At step 812 the protected party opens the question and, at step 814, may answer, decline, or ignore it. Declining and ignoring look the same to the restrained party, who learns only, from the read receipt, that the question was opened. Where the protected party answers, the protected party may add a correction in free text and chooses, at step 816, where the answer and the correction go: the answer, signed by the protected party's client, goes to the restrained party (step 818) and to the protected party's own record; the correction goes to the restrained party, to the decision-maker inbox 710 (step 820), to both, or to neither. Where an answer to Q1 or Q2 shows that the investigator's summary misstates the allegation, and the protected party consents, the ombudsperson sends the decision-maker a discrepancy flag 822 stating only that the protected party has indicated a misstatement, so that the record can be corrected through the ordinary process.

5.8 Evidentiary status and access

[0122] The issuing authority may adopt one of two rules, published to both parties before any submission and stated by each client. Under the first, the issuing authority does not use submissions to the channel, including apologies and the restrained party's question selections, against the

restrained party in its own proceeding, and supports their exclusion from other proceedings to the extent the applicable law allows, so that an apology may be sincere without being treated as a confession. Under the second, an opened apology may be offered by the protected party at the protected party's election, and the restrained party is warned of this before writing. Under either rule investigators have no access to the channel through the system, and the service is configured to release anything in it only in response to legal process, with notice to both parties where permitted. Nothing in this disclosure represents the legal effect of any submission in any proceeding.

5.9 The life of a submission (FIG. 9)

[0123] Referring to FIG. 9, a state machine 900 governs each submission, which passes through the states draft 902, submitted 906, cooling-off 904, screening 908, and then returned for revision 910 (back to draft), rejected 912, or sealed and held 914. A held submission becomes opened 916, expired 918, or destroyed unopened 920 under the protected party's settings, and may carry a withdrawal mark 922. Where screening proposes changes, the submission passes through a state of changes proposed 911, in which it awaits the restrained party's approval, before it is sealed. The restrained party receives the transmittal record 930 upon sealing, a read receipt 932 upon opening 916, and a closure notice 934 upon expiry 918 or destruction unopened 920.

5.10 Safeguards and independence

[0124] The protected party's safety governs every rule. Repeated rejected submissions are reported to the issuing authority as possible harassment under its rules. Rate limits apply per party. The ombudsperson is preferably independent as described in section 1; it may be shared among institutions, and it may be any authority entrusted with the role, such as a court-appointed office or a victim-services office for orders issued by a court. The ombudsperson publishes aggregate statistics (the number of submissions, the share accepted, returned, and rejected, and median screening times), never content. The restrained party may receive confidential counseling through the counselor 706 whether or not anything is submitted.

[0125] The case registry and submission formats may follow an open data standard, so that a channel can follow an order across institutions, or across states under an interstate compact, and so that the same channel can serve orders of schools, employers, and courts.

5.11 Verifiability independent of the intermediary (FIG. 11)

[0126] Each party may wish to understand the other party's mind, meaning what the other actually wrote, read, and answered, without depending entirely on the account of an intermediary. An

ombudsperson or issuing authority that is careless, captured, or overbearing could otherwise alter, reword, delay, or suppress a submission while telling each party a different story, in the manner of a man-in-the-middle. The channel is therefore built so that the intermediary can screen, but cannot impersonate either party, alter a party's words without marking the alteration, or suppress a submission without the suppression becoming visible.

[0127] Referring to FIG. 11, each party's client generates a signing key on the party's own device; the intermediary never holds it. Before submission, the restrained party's client divides the text into segments 1102, for example sentences, computes a salted hash 1104 of each segment, forms a hash tree 1100 over the salted hashes, and signs its root 1106. The screening decision identifies each segment as transmitted, withheld, or replaced. The transmitted version carries, for each transmitted segment 1108, the segment, its salt, and its proof of membership in the tree; for each withheld segment 1110, only its position, its salted hash, and the criterion and reason; and, for each replaced segment 1112, the replacement text signed by the ombudsperson and marked as the ombudsperson's words, at the position of the segment it replaces. The restrained party's approval of the transmitted version is itself signed. At step 1114 the protected party's client verifies that every transmitted segment belongs to the tree whose root the restrained party signed, so that the protected party knows the words shown are the restrained party's own and sees exactly where, and how much, was withheld or replaced, without seeing withheld words. Read receipts and answers are signed by the protected party's client in the same way, so that the restrained party can verify them.

[0128] Each signed root, screening decision, sealing, read receipt, and closure is entered, as a commitment without content, in an append-only witnessed log 712 of the kind used for certificate transparency, whose entries are countersigned by one or more witnesses 714 independent of the ombudsperson and of the issuing authority, for example another institution's ombudsperson, a court clerk, a nonprofit, or each party's advocate. Each client checks inclusion proofs and the log's consistency. Because every submission addressed to a protected party appears in the log, the protected party's client can show, if the protected party wishes, how many submissions were made, how many were transmitted unchanged, transmitted with changes, or rejected in full, and the stated reasons, so that no submission can be silently suppressed; and the restrained party's client can confirm that its submission was logged, what was transmitted, and, from the signed read receipt, that it was opened.

[0129] The clients' code may be published and built reproducibly, with attested builds as described in section 7A of the '657 application, so that a party can confirm that the client performs these

checks and is not controlled by the intermediary. A party who believes a passage was withheld or replaced without basis may ask a reviewer independent of the ombudsperson to examine the decision against the logged commitments. After the no-contact order ends, the protected party may elect to receive withheld passages, which the restrained party's client can release with their salts so that the protected party can verify them against the signed root.

5.12 Privacy-preserving distancing (FIG. 12)

[0130] Purpose. Parties to a no-contact order often share a campus, a workplace, or a neighborhood. Where the protected party requests it and the issuing authority or the ombudsperson finds it necessary, or where both parties elect it, a distancing module 1200 on each party's device helps the two avoid running into each other, by gently rerouting one or both around the other when they are headed for the same place at the same time. It does so without either party, the ombudsperson, or a server learning where the other party is, without security officers following either party, and with as little disruption as possible to the plans and lives of both. Incidental encounters remain possible and are not treated as violations unless the order or the applicable law so provides. The module is designed to be only as heavy-handed as necessary. Its degree of separation is chosen from an ordered set of levels, for example: the channel of section 5 alone; near-field nudges only; nudges plus protected zones; plan-level and schedule-based avoidance; and, at the most, shared-space timetables and administrative accommodations. It starts at the least restrictive level that meets the need stated by the protected party, rises only on a request and a finding that more is necessary, and, where a level has produced no conflicts for a configured period, proposes stepping down at the next periodic review.

[0131] Pairing without disclosure. At issuance, the parties' devices establish a pair secret 1202, for example by a key exchange relayed through the service so that the service never learns the secret, or by derivation from keys exchanged under the channel of section 5. The secret is destroyed when the order ends or distancing is withdrawn.

[0132] Near-field avoidance. Each device broadcasts, at short intervals, rotating proximity tokens 1204 derived from the pair secret and the current time, in the manner of exposure-notification beacons, which bystanders cannot link to a person or to one another and which only the other party's device can recognize. When a device recognizes the other party's token at a signal strength, or an ultra-wideband range, indicating that the other party is within a configured distance (step 1206), it presents a gentle nudge 1208 on that party's own device, for example "You may want to take the other aisle" or "Consider coming back in twenty minutes," without saying where the other party is,

and the other device may do likewise. Recognition occurs only when the parties are already near each other, so a nudge reveals little that sight would not, and no record of the encounter leaves either device. The configuration may nudge the restrained party first, the protected party first, or both. Before the parties come within Bluetooth or ultra-wideband range, the devices can also recognize that they are within a wider, GPS-scale distance of each other, for example a few hundred meters, by private proximity testing: each device, from its own satellite or network position, computes the coarse cell it occupies and the adjoining cells, salts them with the pair secret and the current time window, and the two devices learn, through the private set intersection service 1216 or directly over an encrypted relay, only whether any cell is shared, in the manner of the private proximity testing protocols described in the location-privacy literature. Neither device learns the other's position, and the service learns neither position nor the result. The response is tiered: GPS-scale proximity produces, on the device of the party who is to adjust, a suggestion to change route or timing before the parties are in sight of each other, and Bluetooth-scale proximity produces the closer-range nudge described above. The tests run more often when a device is moving toward the other's last shared cell and rarely when both are stationary, to save battery and limit how often positions are touched.

[0133] Protected zones by oblivious lookup. For places where an encounter would be most distressing, such as the protected party's residence, workplace, or classroom building, or a place named in the order, protected zones 1210 are registered as coarse cells transformed under a keyed pseudorandom function held by an oblivious lookup service 1212. The restrained party's device learns whether its own current or planned cell lies in a protected zone by an oblivious pseudorandom function evaluation, in which the service learns nothing about the cell queried and the device learns nothing about any other cell. Queries are rate-limited and accepted only for cells consistent with the device's own recent movement, so that the restrained party cannot enumerate cells to discover the protected party's zones. A match produces a nudge to route around the zone, the route being computed on the device.

[0134] Plan-level avoidance. Where both parties elect it, each device may derive, from that party's own planned routes, destinations, and calendar, a set of space-time cells 1214 (for example, a coarse area during a fifteen-minute slot) for the coming hours. The two devices learn, through a private set intersection service 1216, only whether their sets intersect and, if so, which of each party's own cells are affected; neither learns the other's other cells, and the service learns neither. On an intersection, the device of the party who is to adjust, chosen under the priority rule of paragraph [0136], proposes

small changes, such as a different entrance, a different route, or a time shifted by a few minutes, ranked by how little they disturb that party's plans.

[0135] Resistance to probing. So that a nudge cannot serve as a locator, the module adds decoy nudges 1218 at a calibrated rate, so that a single nudge does not confirm the other party's presence; limits how many queries a device may make and how finely cells may be drawn; counts only plans consistent with the device's subsequent movement; and has a probing detector 1224 that flags to the ombudsperson, without any location, a pattern consistent with probing, such as repeated plans never acted on or repeated approaches after nudges. No device reports either party's location to the service, the ombudsperson, or the other party.

[0136] Fairness in a shared neighborhood. Distancing should not drive either party out of a neighborhood both share. The module keeps a disruption budget 1222 for each party, measured by the added travel time and the number of plan changes proposed to that party, and balances it between the parties under a priority rule set by the issuing authority or agreed by the parties: for example, the plan made earlier takes priority; the protected party's plans take priority at protected zones and the parties alternate elsewhere; or each party's established routine, such as classes, shifts, or worship, takes priority over discretionary trips. For a place both use regularly, such as a dining hall, gym, library, or house of worship, the ombudsperson may set a shared-space timetable 1220 giving each party alternating windows, each party being told only that party's own windows.

[0137] No officers, and a record only by election. The module works between the parties' own devices and needs no officer to watch or follow either party. Where a restrained party repeatedly ignores nudges and continues to approach, the protected party's device may keep, on that device only, a local evidence log 1226 of the token detections and their times, which the protected party may elect to submit; an automatic report goes to the issuing authority only where the order and the applicable law so provide. When the order ends, the pair secret, zones, and logs are destroyed, except what the protected party elected to keep.

[0138] Schedules and calendars. With each party's consent, the distancing module 1200 connects, through calendar connectors 1228, to the sources that already describe that party's week: a school's class schedule and registration system, an employer's shift or scheduling system, and personal calendars in standard formats such as iCalendar or the interfaces of common calendar services. The schedule is imported to the party's own device, where recurring classes, shifts, practices, and appointments are expanded into space-time cells for the coming days and weeks, with places resolved to buildings and, where a campus or building map allows, to rooms. These cells enter the

private set intersection of paragraph [0134] like any other plan, so that neither party's schedule is disclosed to the other or to the service.

[0139] Recognizing conflicts early. A conflict forecaster 1230 on each device looks ahead, for example each night for the coming week and each term for recurring events, and classifies each intersection as recurring (the same class section, shift, practice, or regular gym hour), one-off (an event, a lecture, a party), or in transit (paths crossing between places). It scores each by the length of the overlap, how close the cells are (the same room rather than the same building), how often it recurs, and the comfort setting each party has chosen, so that conflicts that would be hard to navigate in the moment are resolved days or weeks before they arise. A recurring conflict may be resolved by an accommodation request 1232 that the device of the party who is to adjust, under the priority rule and disruption budget of paragraph [0136], sends through the ombudsperson to the registrar, employer, or other administrator: a section change, a shift swap, a different entrance or seat, or remote attendance for a session. The administrator receives only the accommodation requested, without the other party's identity or schedule and, where policy permits, without the reason; and the change is presented, to the party and to anyone else, as an ordinary schedule change. One-off and transit conflicts produce the gentle suggestions of paragraphs [0132] to [0134]. Wording throughout is neutral ("A schedule adjustment is available for Thursday"), mentions the other party only if the user asks why, and is shown only on the user's own device.

[0140] Periodic review of the degree of separation. At intervals set by the order or agreed by the parties, for example every 30, 60, or 90 days, and whenever either party asks, each party privately reviews the separation through a periodic review 1234 and chooses to keep it, to ask for more distance, or to ask for less. A request for more distance from either party takes effect, within the order's terms, without disclosing which party asked, so that the more protective preference governs. A reduction takes effect only by blind mutual consent 1236: each party's choice is sealed, for example by a commitment or a secure two-party comparison, and the reduction is revealed and applied only if both parties chose less distance; otherwise neither party learns the other's choice, so that a party who chose less is never exposed to the other. Reductions are graduated (for example, smaller protected zones, then nudges only at close range, then the end of plan-level avoidance), each step may be reviewed by the ombudsperson or a counselor for safety, and either party may revert to the previous degree at any time. The module never changes the order itself; where both parties choose less distance, the ombudsperson may tell the issuing authority, at both parties' election, so that it can consider modifying the order.

[0141] Safety and health. The issuing authority may withhold reductions, or the periodic review itself, where the order rests on findings of violence or where the probing detector 1224 has flagged a pattern. Each review offers each party resources, such as counseling and the channel of section 5, and an optional private check on how the party is doing. An apology, an answer, or a structured response in the channel may inform a party's choice but never triggers a reduction by itself. Where the parties choose to restore contact, whether out of forgiveness, reconciliation, or simple practicality, the system gets out of the way as far as the order and the applicable law permit.

6. Privacy, security, and threat model

[0142] The three mechanisms handle some of the most sensitive data a person has: where he, she, or they sleep, with whom, after how much drinking, and what was said after harm. The design follows the privacy principles of the '657 application: processing on the device by default; no server call that the data-flow manifest does not declare; the least data each function needs; encryption in transit and at rest; published retention periods; and code open to inspection and attested builds, as described in section 7A of that application.

[0143] The design addresses the following threats.

- (a) A coercive partner who demands to see the user's phone. The check-in text is neutral or disguised; the duress answer looks like an all-clear; refuge requests and alerts leave no notification visible on the lock screen unless the user chooses; and the client can be locked separately from the phone.
- (b) A person seeking to find a requester at a refuge. The refuge inventory and host addresses are never exposed through any interface available to users; the service answers no inquiry about a person's whereabouts; and credentials, rides, and routes are bound to the requester's device.
- (c) A host who behaves badly, or a guest who does. Hosts are vetted and inspected; the guest room has an inside lock and a help button; either party can summon a responder; and concerns are reviewed and can end a host's eligibility.
- (d) A restrained party seeking to learn about the protected party beyond what receipts disclose. The protected party's settings are not disclosed; read receipts and closure notices state only opening or closure and may be released in batches; and, in the alternative mode of section 5.5, uniform acknowledgments replace them.

- (e) An insider at the ombudsperson service. Plaintext exists only during screening and is destroyed when the submission is sealed; keys are held by the key holder 704, not by the service; and every access is logged.
- (f) An investigator or opposing counsel seeking the channel's contents. The access boundary 152, the sealed store, and the evidentiary rule of section 5.8 apply, and the service releases content only in response to legal process, with notice where permitted.
- (g) A breach of a service. Declarations used in host matching are stored encrypted and evaluated in a trusted execution environment; the refuge service keeps minimal records; and the sealed store holds only ciphertext.
- (h) An intermediary acting as a man-in-the-middle. Signing keys held only on the parties' devices, segment commitments, signed read receipts and answers, and the witnessed log of section 5.11 let each party verify what the other wrote and read, detect any unmarked alteration or any suppression, and ask an independent reviewer to examine any screening decision.
- (i) A party seeking to use distancing to locate the other. Tokens are recognizable only near the other party; zone lookups are oblivious, rate-limited, and tied to the device's own movement; plan cells are coarse and compared by private set intersection; decoy nudges mask true ones; and no location is sent to anyone.

7. Computing device (FIG. 10)

[0144] Referring to FIG. 10, a computing device 1000 on which a client or service of this disclosure, or a part of one, executes includes a processor 1002; memory 1004 holding instructions that, when executed by the processor, perform the methods described; storage 1006; a secure element or trusted execution environment 1008 for keys, credentials, and encrypted evaluation; wireless interfaces 1010, which may include cellular, Wi-Fi, Bluetooth Low Energy, ultra-wideband, and near-field communication; a satellite positioning receiver 1012; sensors 1014, which may include an inertial measurement unit, a barometric pressure sensor, an ambient-light sensor, and, through a local link, a breath-alcohol sensor; a display 1016; a haptic actuator 1018; an optional microphone and speaker 1020; and a power source 1022. A service may run on one or more such devices in a data center, and a client on a phone, watch, ring, tablet, room hub, or computer. The devices of FIG. 9 of the '657 application are examples.

8. General

[0145] The examples, values, and thresholds given here (for example, dwell periods, time windows, breath-alcohol levels, cooling-off periods, and rate limits) are illustrative and may be set by a user, an operator, or a policy. A step described as performed by one component may be performed by another, or divided among several. A feature described in one section may be combined with any feature of another section or of the '657 application. Where this disclosure describes a person by "he, she, or they," or by a role, it applies equally to persons of every gender and to every relationship form. Nothing in this disclosure requires any person to use the system, and declining any feature carries no inference about the person who declines it.

[0146] The words "comprising," "including," and "having" are open-ended. "A" or "an" means one or more. "Or" is inclusive unless the context requires otherwise. A claim to a method also discloses a system configured to perform it and a non-transitory computer-readable medium storing instructions that cause a processor to perform it, and the reverse.

9. Bedroom guard state and related embodiments (FIG. 13)

[0147] Covert arming of a bedroom guard state. A user may enroll, as an arming cue, a media selection (a playlist, a track, an artist, or a station), an ordinary request to a voice assistant, or an ordinary command to a connected device. When the user device, such as a phone or smart speaker, is determined to be in the user's bedroom, from its assignment to a bedroom in a smart-home application, a room tag of the bedroom, or a designation learned from where the device rests during the user's sleep periods, and the enrolled cue is given by voice, by touch, or through a media-control interface without any command naming a safety function, the device carries out the ordinary request (for a media selection, it plays the selection) and enters a guard state. Entering the guard state produces no output other than the ordinary response and a private acknowledgement, such as a haptic pulse, on a device of the user. In the guard state no affirmation is awaited and no playback is conditioned on any affirmation, and the guard state never implies consent to anything. The device detects, on the device and without first detecting a wake word, a safe phrase in its microphone audio, from which any media it is playing is cancelled using that media as a reference, and, by default, it retains no audio and generates no transcript of speech that does not match a safe phrase; where the user's permissions and settings and applicable law so provide, the device may instead record audio upon a safe phrase or other sign of a consent conflict in the bedroom. Any such recording is off by default, is encrypted to a key held only by the user, is never made available to the other person, to a platform, or to the provider of the system without the user's election, and is

made only where the law applicable to recording the other person permits it, including any requirement that every party consent. Upon a safe phrase, the device stops any playback and runs an escalation of a private check-in on a device of the user, then an alarm, then a call for help through a pathway the user configured, each stage advancing only upon absence of a response from the user. The guard state continues until closed by the user who armed it. This embodiment may be combined with the guard state, bedroom recognition, and safe-word registry described in U.S. Provisional Patent Application No. 64/183,657.

[0148] Distancing after a block on a dating or social platform. The privacy-preserving distancing of section 5.12 applies equally where no order exists and one user of a dating, social, or messaging platform has blocked another. Upon the block, the platform, or the two users' clients without the platform's involvement, establishes a pair secret known to the two clients and to no server, and each client broadcasts rotating proximity tokens derived from the pair secret and the current time, unlinkable by any device that does not hold the pair secret. When one client recognizes the other's token, or determines by private proximity testing over coarse location cells salted with the pair secret that the two are near each other or that a venue or event either has planned is shared, it presents to its own user a suggestion to change route, venue, or timing, without indicating the other user's location, identity, or presence, and the platform receives no location of either user. The blocking user alone may elect whether the blocked user also receives suggestions, so that the blocked user can avoid an encounter without learning that the blocking user is near. Decoy suggestions at a calibrated rate keep a single suggestion from confirming the other's presence. Either party's client may end its own participation, and unblocking ends the pair secret.

[0149] Guard state shared across a user's devices. The guard state of [0147] may be armed on any one device of the user, by an arming cue, by an explicit command, or automatically where the user has so elected upon detection, by short-range wireless detection, of a device of a second person, other than a device of the user, in the user's bedroom. The detection evaluates short-range wireless signals within short windows on the user's own device, and no identifier of the second person's device is stored or transmitted. The arming device then sends, over a local link and authenticated by keys shared among the user's devices, an arming message to the user's other devices present in the bedroom, such as a phone, a watch, earbuds, a smart speaker, a smart display, a television, and a smart lock, each of which enters the guard state, honors the safe phrase, the brake, concern cues, and button or touch patterns of the user, and produces no change apparent to the second person. A safe phrase or other input detected by any of these devices is propagated to the others, so that one

device stops its playback, another delivers the private check-in, a third sounds the alarm, and the one with the best connection places the call for help, each according to its capabilities and the user's settings. The guard state ends on all of the devices only when the user who armed it closes it.

[0150] Inferring that a room is the user's bedroom. A device may determine that it is in the user's bedroom without any manual setup, by learning on the device, over a number of nights, where the user sleeps. Signals that indicate a sleep period include: the user's phone or watch being connected to a charger for several hours during night hours; the device's own record of the user's sleep and wake cycle, such as inactivity of the phone and a sleep or do-not-disturb mode; a wake-up alarm set at night and turned off in the morning after an interval consistent with sleep; sleep stages, heart rate, and stillness sensed by a wearable; and the recurring acoustic event of snoring, of which only a count or flag is kept and no audio. During each such period the device records a fingerprint of the place where it rests, comprising at least one of the signal strengths of nearby wireless networks and devices, a barometric pressure indicating the floor, a local magnetic field, and the coarse location of the dwelling. A place whose fingerprint recurs over a configured number of sleep periods is proposed to the user as a bedroom and is designated as one upon the user's confirmation. A fixed device such as a smart speaker may adopt the designation when the user's phone or watch is nearest to it during the user's sleep periods, as determined by short-range wireless or acoustic ranging. Thereafter the device determines that it is in the bedroom when its current surroundings match the fingerprint, optionally together with a bedtime signal, such as charging after a configured time of night, activation of a sleep mode, setting of a wake-up alarm, or a bedtime routine of a voice assistant or smart-home system. A separate designation may be learned for each user of a shared device, and a designation is never applied to a user who has not been established as an adult.

[0151] Labeling and probabilistic recognition of the bedroom. The user may simply label a place, a room, or a fixed device as the user's bedroom in the application, with no learning at all, and the label takes effect at once. Absent a label, the device may compute, on the device, a probability that it is in the user's bedroom by combining a plurality of signals, each weighted by how strongly it indicates a bedroom, for example in a naive Bayesian or logistic model whose weights are learned or set in advance. The signals include those of [0150] and further: a name given to a device or room in a smart-home application, such as "bedroom" or "master"; assignment of a fixed device to a bedroom group; a recurring night-time scene or routine of a smart-home system, such as "good night," or lights in the room being switched off at night; playback of sleep sounds, white noise, or a sleep playlist in the room at night; occupancy reported by a smart bed or mattress sensor; the phone

lying flat and motionless for several hours at night, as reported by its accelerometer and gyroscope; low ambient light and a low ambient noise level, as reported by its light sensor and by a sound-level reading that keeps no audio; and a non-imaging presence sensor, such as a millimeter-wave radar, reporting that a person has lain still for several hours. A fixed device such as a smart speaker or smart display may further use: the acoustic response of the room, measured by playing a test signal through its loudspeaker and recording the reflections with its microphones as some speakers already do to tune their sound, a short and highly damped reverberation suggesting a small room with a bed and soft furnishings; its own pattern of use, such as alarms and sleep timers set on it, night-time and morning routines run through it, and morning briefings requested of it; an absence of speech at the device through night hours followed by the first speech in the morning; the rooms or names already assigned to the household's other fixed devices, such as kitchen and living room, which leave the remaining device, particularly on an upper floor, likely to be in a bedroom; and its ambient light sensor reporting darkness for long periods at night. Each such signal is taken as one input to the probability, none being treated as conclusive alone. Several of these signals require no location, Bluetooth, health, or microphone permission, so that the device may recognize a bedroom from the time of night, the charging state, the device's motion and orientation, ambient light, barometric pressure, and the user's own label alone. When the probability exceeds a threshold, the device proposes the designation to the user, or applies it where the user has so elected, and the safe-word sensitivity or guard state of [0147] may then follow.

[0152] Protection matched to the stage of a relationship. Where it is known that two users are early in, or just starting, a relationship, for example because each has so indicated, because the users matched on a linked dating platform within a configured period, or because they have not previously designated each other as established partners, the device applies the affirmative consent envelope of the '657 application during an intimate session: playback continues only while each participant renews affirmation within the affirmation window, and upon a lapse the playback fades naturally and a check-in is issued. As the relationship matures, the participants may each elect that the envelope no longer be applied between them, whereupon the device operates in the guard state of [0147], in which no affirmation is awaited and no playback is gated. The envelope is withdrawn only when every participant has made that election privately on his, her, or their own device; no participant's election is disclosed to the other; any participant may restore the envelope at any time and for any session, without giving a reason; and the change never implies consent to anything. In every stage the safe phrase, the brake, and the escalation to help remain armed.

[0153] Recommending a setting for the envelope. At any time, including at the start of a session, during a session, or in a later check-in such as a morning-after check delivered privately the following day, the device may recommend a setting for the affirmative consent envelope of [0152] and privately ask each participant, on his, her, or their own device, to affirm or decline it. The available settings include keeping the envelope as configured; moving its phaseout point, meaning the point after which the envelope gives way to the guard state of [0147], earlier or later, the point being defined by an elapsed time, by a stage of a session, or by a number of renewals; and enlarging the affirmation window so that it spans an entire session from its start, so that the opening affirmation of each participant carries that session and no renewal is requested unless a participant restores renewal. The device forms its recommendation from signals such as the stage of the relationship described in [0152], the time since the participants matched or designated each other as established partners, each participant's own stated preferences and earlier choices, and recent events. A restore of the envelope by any participant, or a stop, concern cue, or call for help in a recent session, leads the device to recommend a more protective setting for a period and never a less protective one, and the device never presents a stop or a concern as a cost. Preferably, a recommendation that would ask for fewer renewals is presented outside an intimate session, for example in the morning-after check, when each participant can consider it calmly, while a recommendation for more renewals may be presented at any time. In duo mode (also called duet mode), in which every participant is enrolled and aware of the system, a setting that asks for fewer renewals takes effect only when every participant has affirmed it privately; where the responses differ, the most protective setting applies, and no participant learns another's response. A setting that asks for more renewals takes effect on the affirmation of any one participant. In solo mode, in which only the user is enrolled, the user's own affirmation suffices, because the envelope then governs only the prompts the user receives; a stop word spoken by any person present, including a partner who is not enrolled, is still honored. Under every setting the safe phrase, the stop word, the brake, and the escalation to help remain armed; any participant may shorten the window or restore the envelope at any moment without giving a reason; the participants may optionally keep a single check-in before any stage they marked in advance; and no setting, and no affirmation of a setting or at the start of a session, implies consent to anything.

[0154] Keeping recommendations effortless. The recommendations of [0153] are designed to need as little attention as possible, so that the envelope does not become a chore. The device recommends a change rarely, at most once in a configurable period, and preferably folds the recommendation

into a touchpoint the participants already see, such as the morning-after check, rather than adding a prompt of its own. Each recommendation arrives with the suggested setting already selected and a one-line reason, and is accepted by a single input: a tap, a press of a button, a squeeze or other enrolled touch on a wearable device, or a short spoken word such as "okay." An edit option opens the setting with the suggestion filled in, so that a participant who wants something different changes only what differs. A participant may elect automatic acceptance, in which case a setting that keeps or increases protection is applied with only a private notice. In duo mode a setting that asks for fewer renewals still takes effect only on each participant's own affirmation under [0153], though a single input suffices and several pending changes may be accepted together; in solo mode, where the envelope governs only the user's own prompts, automatic acceptance may extend to every recommended setting. Absence of a response leaves the current setting unchanged and is never treated as acceptance of a less protective setting. A participant may also dismiss recommendations of a given kind for a period.

[0155] Variations of the guard state. The guard state of [0147] and [0149] is not limited to the combination described there, and each of its elements may be used, omitted, or replaced independently of the others. The arming cue may be any ordinary input that does not name a safety function, including a media selection, a request to a voice assistant, a command to a connected device, a press or touch pattern on a button, a remote, or a wearable device, a gesture, or placing a phone in a particular position or orientation; it may be designated by the user or supplied by a default configuration, such as a bedtime routine; and an explicit command naming the guard state may also arm it. Any ordinary response the cue has is carried out as usual. In the guard state the device responds not only to the safe phrase but also, in place of it or in addition to it, to a non-verbal safe input of the user, such as a tap or touch pattern on a watch, phone, or remote, a gesture, or a button pattern, which starts the same private check-in and help response as the safe phrase. Where the arming cue is a media selection, the media it starts may itself serve as an ambient signal: upon a safe phrase, the device may, at the user's election, fade that media to a reduced level rather than stop it, so that the fade marks to everyone in the room that a check-in is pending, and resume it upon an all-clear answer, while continuing to cancel that media from the microphone audio; where the user has made no election, the silent private check-in without any change to the media remains the default. Where the participants have elected the affirmative consent envelope of [0152], the media selection that armed the guard state may also open the envelope, continuing to play only while each participant renews an affirmation within the affirmation window, a lapse fading the media selection

and issuing a check-in. Within the envelope, the volume of the media may follow the current consent grade, computed preferably from the tier of each participant's enrolled go words rather than from vocal tone, so that the music itself is an ambient indication of where the grade stands: for example, somewhat stronger at a high grade such as A+, moderate at a middle grade such as A, soft at a lower grade such as A-, and faded out, with a check-in, at a grade such as B. The mapping from consent score to volume may be any function, linear or nonlinear and preferably monotonic, such as a set of steps, a logarithmic or perceptually weighted curve, a sigmoid, or a curve with a threshold below which the media fades out, and the score may be a letter grade, a numeric score such as that produced by the consent scoring of the '301 patent, or a tier of go words. The consent score is not fixed for a session, and affirmative consent is not left to coast: absent a deliberate affirmative or enthusiastic signal from a participant, through any input that participant enrolled, such as a go word, a gesture, a squeeze or touch pattern, or a button, the score decays gradually over time, at a rate or along a curve the participants set, so that the volume drifts lower as an ambient prompt for feedback. A renewed signal restores the score, and a score that decays below the threshold fades the media and issues a check-in. Involuntary physiological signals, such as heart rate or signs of arousal, never raise the score. The volume follows the lowest current grade among the participants, so that one participant's enthusiasm never masks another's hesitation; it changes smoothly, within a range and a ceiling the participants set; the mapping is used only where every participant has elected it; and a lower grade or a fade is never treated as a fault of anyone. The guard state may be entered in any room, or only while the device is determined to be in the user's bedroom. The steps may be performed by a single device or divided among the user's phone, wearable devices, smart speakers, and a server, and the safe phrase may be detected on the device or, where the user permits, by a service in communication with it, in either case without first detecting a wake word. Cancelling the device's own playback, stopping playback upon a safe phrase, producing no perceptible output on arming, and awaiting no affirmation are each preferred but optional; for example, a device may show a status indicator that its platform or the law requires, or may duck rather than cancel its playback. Upon a safe phrase the device presents a private check-in and, upon absence of a response, initiates a help response, which may comprise one or more of notifying a contact the user designated, connecting to a monitoring center, sounding an alarm, and requesting emergency services through a pathway configured by the user or by default, in an order and with timings that the user or an applicable policy sets.

[0156] Limiting false alarms. Because a check-in can go unanswered for innocent reasons, such as inattention, sleep, or a device out of reach, the guard state is designed to verify before it summons emergency services. First, a safe phrase is accepted only when its detection confidence exceeds a threshold and, preferably, when it is repeated within a short interval or matches the user's enrolled voice; the device's own media is cancelled, and playlists may be screened for the phrase, so that a song cannot trigger it. Second, the private check-in is delivered at once on several of the user's devices, such as a haptic pulse on a watch, a notification on a phone, and, unless the user chose discretion, a spoken prompt, and is repeated once with greater intensity; it is answered by a single input, such as a tap, a squeeze, or a short phrase in the user's verified voice, and a duress answer that looks like an all-clear answer initiates the help response. Third, the help response is tiered: it first notifies a contact the user designated, who may call back or check in, and optionally a monitoring center that attempts twice to reach the user; it requests emergency services only upon an answer requesting help, a duress answer, a corroborating distress signal (such as a scream, an impact, a pattern consistent with freezing, or a physiological distress signal from a wearable device), or a failure to reach any designated contact within a configured period. Fourth, a request to emergency services carries a validation level computed from the evidence, distinguishing, for example, a safe phrase followed only by silence from a safe phrase accompanied by a scream or by an answer requesting help, in the manner of the alarm-validation levels used by monitoring centers, so that a dispatch center can set its response accordingly. Finally, the device learns from detections that the user dismisses as mistaken, prompting the user to choose a less ambiguous safe phrase, and never lowers protection after a detection that the user did not dismiss. Where a safe phrase has been detected and the user then cannot respond, the absence of a response is itself a reason for concern, so these measures verify the escalation rather than suppress it.

[0157] Guard state away from the user's own speaker. The guard state works when the user is away from home, for example at a new partner's residence whose speaker comes from a different manufacturer and plays from a music service with which the user has no account. The user's own phone, watch, or earbuds may perform every step: the arming cue may be a tap or gesture on the watch, a placement of the phone, or an ordinary request that the user speaks to the host's speaker and that the user's own device recognizes, on the device, as the user's enrolled cue. Where the enrolled cue is a media selection played by another device or service, the user's device may recognize the selection by its sound, by matching an acoustic fingerprint of the ambient audio against a fingerprint of the enrolled selection computed on the device, and arm upon recognition,

whatever device or service plays it. To listen for the safe phrase over that music, the user's device may identify the track and its position by the same fingerprint, obtain the corresponding audio from a catalog or from the user's own music service, align it with the ambient audio, and use it as the reference for cancellation; where no reference can be obtained, the device may suppress the music by source separation or rely on the microphone of a wearable device nearer the user's mouth. In addition, a device that detects the universal emergency phrase but is not bound to the user, such as the host's speaker, may broadcast over a short-range wireless link an anonymous alert that the phrase was heard, carrying no identifier of any person or account and sent under a rotating broadcast identifier. Any nearby device whose user has the guard state active, of whatever manufacturer, treats the alert as a detection of the safe phrase and starts its own private check-in and help response, so that help reaches the user's own contacts rather than anyone the host's device knows. The alert format may be published as an open standard so that devices of different manufacturers can send and receive it, and the broadcasting device takes no further action of its own beyond any local alarm its owner has configured. A nearby device on which no guard state has been armed, and which has no safety application beyond its operating system, may also act on the alert: its operating system may present to its user a private check-in offering an option to request help, which, if chosen, notifies the emergency contacts and places the emergency call configured in the operating system. Because every device in range of the broadcast, including the devices of neighbors and other guests, may receive the alert, such a device initiates no help response upon absence of a response, so that the alert reaches a person who needs help without causing everyone nearby to summon it.

[0158] Keeping media from triggering the guard state. Wherever this disclosure refers to cancelling media from the microphone audio, any one or more of the following techniques may be used in place of cancellation or in addition to it, on the user's device, on another device in the room, or, where the user permits, by a service in communication with it. (a) Cancellation, using as the reference the media that the device itself plays, or audio of the identified track obtained from a catalog or a music service and aligned with the captured audio. (b) Lyric matching: a candidate detection of a safe phrase is rejected where its words coincide in time and content with the lyrics or transcript of the track being played, obtained from the track's metadata, from a lyric service, or by transcribing the reference audio on the device. (c) Source classification: a candidate detection is rejected where it is classified as reproduced by a loudspeaker rather than spoken live in the room, from acoustic features such as the frequency response and band limits of a loudspeaker, the

dynamic-range compression of a mastered recording, steady or pitch-corrected sung pitch, a steady beat or accompanying instruments, the reverberation of a recording space added to that of the room, coding artifacts, a direction of arrival matching the location of the playing device, and the absence of the breath sounds and near-field effects of a voice close to the microphone. (d) Voice verification: a candidate detection that matches the enrolled voice of the user or of another participant is treated as spoken live. (e) Watermarking: a music service, a playing device, or a publisher embeds an inaudible watermark in the media, throughout it or over the spans whose lyrics contain a safe phrase or a phrase like one, carrying the identity of the track and the playback position or the words of the span; the listening device detects the watermark and uses it to obtain the reference or the lyrics, or to reject a candidate detection whose words match the lyrics of the watermarked span. (f) Separation of the user's voice from the media by source separation, or capture by the microphone of a wearable device nearer the user's mouth. (g) Screening of tracks for the phrase before they play, and skipping, muting, or substituting a segment that contains it. Because a person may call for help while music plays, and because another person may play audio to defeat the guard state, each technique that rejects a candidate detection rejects only that detection and never stops listening for the duration of a track; a watermark suppresses only a candidate detection whose words match the lyrics of the watermarked span, and only where the watermark is authenticated, for example by a signature of the music service or playing device verified on the listening device, so that an unauthenticated or forged watermark cannot silence the guard state; and where a classification is uncertain, the candidate detection is accepted, the verification measures of [0156] guarding against a false alarm. Likewise, where a media selection played by another device or service serves as the arming cue, the user's device may recognize it by an acoustic fingerprint, by a watermark embedded in the media, or by metadata identifying the media being played that the playing device reports or that a casting, shared-listening, or media-control session to which the user's device is connected provides.

EXEMPLARY AND ILLUSTRATIVE CLAIMS

The following claims are exemplary and illustrative. They are presented to illustrate the subject matter disclosed and do not limit the scope of this disclosure or of any claims that may later be presented in an application claiming the benefit of this provisional application.

1. A method performed by a mobile computing device of a user, the method comprising: detecting, on the device, an arrival of the device at a place; determining, on the device, that the place is unfamiliar by comparing the place with a set of known places stored on the device; determining that a time of the arrival falls within a configured late-night window; determining, on the device, that a risk context is present based on at least one signal from a period preceding the arrival, the at least one signal being selected from a breath-alcohol reading at or above a configured level, an on-device inference of a drinking episode, a departure from a drinking occasion, an indication of a date, and co-travel of another device with the device; responsive to the place being unfamiliar, the time falling within the late-night window, and the risk context being present, presenting a private check-in on the device or on a wearable bound to the user, the check-in offering at least an all-clear answer, a greyzone answer, and a help answer; and responsive to the greyzone answer, sending an alert to a contact the user designated in advance.

2. The method of claim 1, wherein the set of known places comprises places labeled by the user, sleeping sites recognized from the user's device signals, and places whose familiarity statistics, comprising a number of distinct days of presence or a cumulative dwell time over a trailing period, exceed a threshold, and wherein a place is unfamiliar when it matches no such entry.

3. The method of claim 1, wherein familiarity is determined at the level of a floor using an estimate of the device's floor from a barometric pressure sensor, such that a floor of a known building may be unfamiliar.

4. The method of claim 1, further comprising classifying, on the device, the place as a residence from at least one of map data cached on the device, characteristics of nearby wireless networks, the time of day, and an estimated floor, and treating a place whose classification is uncertain as a residence, wherein the check-in is presented only for a place classified as a residence.

5. The method of claim 1, wherein co-travel is determined from the persistence of short-range wireless signals of a single nearby device across a trip, evaluated within short windows without retaining identifiers of that device, and excluding devices of the user and of contacts the user designated in advance.

6. The method of claim 1, wherein the check-in further accepts an enrolled duress answer whose visible effect on the device is identical to that of the all-clear answer and which silently invokes a help pathway designated for duress.

7. The method of claim 1, further comprising, responsive to the greyzone answer, offering on the device at least one of: arming a state in which the device detects safe words on the device; a pretext call or message; a ride; and a request for a refuge for the night.

8. The method of claim 1, wherein absence of an answer does not by default raise an alarm, and the user selects in advance among taking no action, repeating the check-in once, notifying the designated contact that the check-in went unanswered, and advancing through an alarm stage and a help stage.

9. The method of claim 1, wherein the check-in is presented without sound and with neutral or user-chosen disguised text, and wherein no output concerning the check-in is produced by any device located at the place other than the user's own devices.

10. The method of claim 1, wherein the set of known places, the at least one signal, and the determination of the risk context remain on the device, and the alert includes only location information the user elected in advance to share.

11. The method of claim 1, further comprising suppressing a later check-in at the same place on the same night after an all-clear answer, and never suppressing a later check-in on a night on which the greyzone answer was given.

12. A method performed by a mobile computing device of a user, the method comprising: determining, on the device, that the device is at the user's own residence, that another device not previously associated with the user has arrived at the residence together with the user's device within a configured late-night window, and that a risk context is present based on at least one signal from a preceding period selected from a breath-alcohol reading at or above a configured level, an on-device inference of a drinking episode, a departure from a drinking occasion, and an indication of a date; and, responsive thereto, presenting on the device a private check-in offering at least an all-clear answer, a greyzone answer, and a help answer, and offering to arm a state in which the device detects safe words on the device.

13. A mobile computing device comprising a processor, a positioning subsystem, and memory storing instructions that, when executed by the processor, cause the device to perform the method of any one of claims 1 to 12.

14. A non-transitory computer-readable medium storing instructions that, when executed by a processor of a mobile computing device, cause the device to perform the method of any one of claims 1 to 12.

15. A method performed by one or more computing devices of a refuge coordination service, the method comprising: receiving a request for a refuge for a night from a device of a requester or by a press pattern, enrolled by the requester, on a button in a room, the request comprising a location of the requester; screening the request for medical risk against a rule set and, where the rule set is met, routing the request to medical evaluation or to a monitored refuge; otherwise identifying candidate refuges from an inventory of safe-sleep rooms and a registry of homes of vetted hosts; selecting a refuge from the candidates; where the selected refuge has one or more access devices, causing an access-control system to issue to the requester's device a time-limited credential for each access device on a path to the selected refuge; and withholding the identity and location of the selected refuge from every person other than the requester and those needed to provide it.

16. The method of claim 15, wherein the request is made by a press pattern on a button mounted at a bed, or by a greyzone answer to a check-in presented by the requester's device upon arrival at an unfamiliar residence.

17. The method of claim 15, wherein selecting a home of a host comprises applying a no-romantic-interest constraint based on declarations, made privately to the service, of the genders of persons toward whom the host and, optionally, the requester may feel attraction, under which a pair of the requester and a host is excluded where either party's declaration indicates possible attraction toward persons of the other party's gender, and, where the requester has made no declaration, the pair is excluded unless the host is of a gender the requester has stated to be acceptable.

18. The method of claim 17, wherein the constraint further excludes a pair having a prior romantic or dating connection, determined by private set intersection between a keyed hash of a pseudonymous identifier of each of the requester and the host and a set of keyed hashes of pseudonymous identifiers of persons with whom the other has matched or corresponded, the service learning only whether an intersection exists.

19. The method of claim 17, wherein the declarations are stored encrypted and evaluated within a trusted execution environment, such that neither the requester nor the host learns any declaration of the other beyond what is implied by the match itself.

20. The method of claim 15, wherein selecting comprises excluding a refuge located in the building or on the floor from which the requester is leaving, or associated with a person the requester names.

21. The method of claim 15, wherein the state of each safe-sleep room in the inventory is reported by its electronic lock and by a non-imaging presence sensor in the room.

22. The method of claim 15, further comprising, where screening indicates possible impairment that does not meet the rule set and the requester consents, monitoring respiration in the refuge with a non-imaging sensor and alerting a responder if respiration is absent or abnormal for a configured period.

23. The method of claim 15, wherein, where the refuge is a host's home, arrival is verified by a short-range handshake between the requester's device and the host's device or by the host's device scanning a code displayed by the requester's device.

24. The method of claim 15, further comprising booking a ride for the requester alone with a destination entered by the service and not displayed on the requester's lock screen, and presenting a check-in upon a deviation of the ride from its route or a long stop.

25. The method of claim 15, further comprising, at a check-out time, revoking the credential and presenting to the requester once an aftercare step listing medical, counseling, and reporting options.

26. The method of claim 15, wherein the service sends no notification to a person the requester left, and offers the requester a neutral message to send at the requester's election.

27. A safe-sleep room comprising: a door having an electronic lock that accepts time-limited credentials and an inside release that unlocks and opens the door in one motion; a bed having a headboard; a help button mounted at the center of the headboard; a room tag giving the room a dispatchable location; a non-imaging presence and respiration sensor; and a hub that reports the button, the sensor, and the lock state to a refuge coordination service; wherein the room has no camera and no microphone.

28. A system comprising one or more processors and memory storing instructions that, when executed, cause the one or more processors to perform the method of any one of claims 15 to 26.

29. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause the one or more processors to perform the method of any one of claims 15 to 26.

30. A method performed by one or more computing devices of an intermediary service entrusted with an intermediary role between parties to a no-contact order, the method comprising: registering

a no-contact order between a restrained party and a protected party; receiving from a client of the restrained party a submission comprising an apology; after a cooling-off period during which the restrained party may revise or withdraw the submission, presenting the submission for screening against published criteria; upon acceptance, encrypting the submission to a key held by a key holder designated by the protected party, recording a hash of the encrypted submission in an audit log, and destroying the plaintext; holding the encrypted submission and notifying the protected party only according to a notification mode the protected party selected from a set comprising notifying of each item, notifying once, and never notifying; returning to the client of the restrained party a transmittal record stating the content transmitted to the protected party and identifying each part of the submission changed or withheld in screening, with a reason for each; and denying investigators of the matter access to the submission.

31. The method of claim 30, further comprising, upon the protected party's client decrypting the submission, returning to the client of the restrained party a read receipt signed with a key held on the protected party's device and identifying the version decrypted, the protected party having been informed before opening that opening sends a read receipt.

32. The method of claim 30, wherein screening comprises proposing to withhold or replace a part of the submission, presenting to the restrained party a comparison of the submission as written and a proposed transmitted version marking each withheld or replaced part and its reason, and sealing the transmitted version only upon the restrained party's approval.

33. The method of claim 30, further comprising: receiving from the client of the restrained party a selection of a clarifying question from a closed set of templates, the set comprising a question whether an investigator's summary of an allegation in the matter is accurate; holding the question as the apology is held; receiving from the protected party an answer, a declination, or no response, wherein a declination and no response produce identical outputs to the restrained party; and distributing an answer to the restrained party and any correction the protected party adds according to the protected party's election among the restrained party, a decision-maker, both, and neither.

34. The method of claim 33, wherein the closed set further comprises a question of scope answered from a list of charges in the matter and a question of preferred outcome answered on a graduated scale of severity and range of response.

35. The method of claim 33, further comprising, where an answer indicates that the investigator's summary misstates the allegation and the protected party consents, sending a decision-maker a flag stating only that a misstatement has been indicated.

36. The method of claim 30, wherein screening comprises flagging by an automated pre-screen of content selected from threats, requests, inducements, blame-shifting, sexual content, information about the protected party's whereabouts, and third-party messages, and a decision by an operator of the intermediary service to accept, return for revision with the unmet criteria stated, or reject, with referral to an issuing authority of the order where the content is threatening or retaliatory.

37. The method of claim 30, wherein the protected party's settings govern only delivery and never prevent the restrained party from making a submission.

38. The method of claim 30, further comprising receiving a withdrawal notice for the apology and marking the held apology as withdrawn by the sender without deleting it, the withdrawal notice being listed in the transmittal record.

39. The method of claim 30, further comprising displaying to the restrained party, before composition, the evidentiary status of submissions under a rule adopted by an issuing authority of the order, and recording in a case registry any designation by the issuing authority of use of the service as authorized communication.

40. The method of claim 30, wherein the order is mutual and each party may submit and each party's settings protect that party.

41. A system comprising one or more processors and memory storing instructions that, when executed, cause the one or more processors to perform the method of any one of claims 30 to 40.

42. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause the one or more processors to perform the method of any one of claims 30 to 40.

43. A method comprising: presenting, by a mobile computing device upon its arrival within a configured late-night window at an unfamiliar residence in a risk context, a private check-in offering graded answers; responsive to a greyzone answer, offering on the device a request for a refuge for the night; and, responsive to acceptance of the offer, transmitting the request to a refuge coordination service that screens the request for medical risk, selects a refuge, and issues to the device a time-limited credential for the refuge.

44. The method of claim 1, wherein detecting the arrival comprises at least one of the device joining a wireless network the device has not previously joined and a barometric pressure sensor of the device indicating a climb of one or more floors following a ride or walk.

45. The method of claim 1, further comprising presenting a second check-in a configured interval after the check-in, independent of any change in the place.

46. The method of claim 1, further comprising learning, on the device, from the user's answers, wherein a place answered with the all-clear answer repeatedly is proposed as a known place, and wherein no learning that suppresses check-ins is performed from a greyzone, duress, or help answer.

47. The method of claim 15, wherein the request is received from a button press alone, the method further comprising sending to the requester's device one-tap screening questions concerning danger and intoxication and, where the questions go unanswered, treating the requester as possibly impaired under the rule set.

48. The method of claim 30, further comprising limiting the restrained party to a configured number of apologies per period, wherein a later apology replaces an earlier apology that has not been opened, the replacement being recorded in the audit log and listed in the transmittal record.

49. The method of claim 30, further comprising enabling the protected party, upon opening the submission, to verify against the hash recorded in the audit log that the opened submission is the submission that was screened.

50. The method of claim 30, wherein the client of the restrained party signs, with a key held on the restrained party's device, a root of a hash tree over salted hashes of segments of the submission, and wherein the client of the protected party verifies, before displaying the transmitted version, that each transmitted segment belongs to the signed tree, and displays the position and stated reason of each withheld segment without its content.

51. The method of claim 50, wherein a segment replaced in screening is displayed as text of the intermediary service, signed by the intermediary service, at the position of the replaced segment.

52. The method of claim 30, further comprising entering commitments, without content, to each submission, screening decision, sealing, and read receipt in an append-only log countersigned by at least one witness independent of the intermediary service and of an issuing authority of the order, and verifying inclusion of the entries at the clients of the parties, whereby suppression of a submission addressed to the protected party is detectable.

53. The method of claim 52, further comprising presenting to the protected party, at the protected party's election, counts of the submissions addressed to the protected party that were transmitted unchanged, transmitted with changes, and rejected in full, with the stated reasons.

54. The method of claim 30, further comprising, where the submission expires or is destroyed without being opened, returning to the client of the restrained party a closure notice stating only that the submission was closed without being opened.

55. The method of claim 30, wherein, in an alternative mode, read receipts and closure notices are withheld and the restrained party receives instead an acknowledgment identical in content and size regardless of the protected party's settings and actions.

56. The method of claim 30, wherein the intermediary service is operated independently of the investigation and adjudication of a matter underlying the order.

57. A method performed by a mobile computing device of a first party to a no-contact order with a second party, the method comprising: holding a pair secret shared with a device of the second party and not known to any server; broadcasting proximity tokens derived from the pair secret and the current time, the tokens being unlinkable by devices that do not hold the pair secret; recognizing a proximity token broadcast by the device of the second party and estimating that the second party is within a configured distance; and, responsive thereto, presenting on the device a suggestion to change direction, route, or timing without indicating the location of the second party; wherein no location of either party is transmitted to a server or to the other party.

58. The method of claim 57, further comprising determining whether a current or planned location cell of the device lies within a protected zone by an oblivious pseudorandom function evaluation with a lookup service that learns nothing of the cell queried, wherein queries are rate-limited and accepted only for cells consistent with the device's recent movement.

59. The method of claim 57, further comprising determining, by private set intersection with the device of the second party, whether planned space-time cells of the parties intersect, and, where they intersect and the first party is to adjust under a priority rule, proposing a change of route, entrance, or time ranked by disruption to the first party's plans.

60. The method of claim 57, further comprising presenting decoy suggestions at a calibrated rate such that a single suggestion does not confirm the presence of the second party.

61. The method of claim 57, further comprising maintaining a disruption budget for each party, measured by added travel time and number of plan changes, and balancing proposed changes between the parties under a priority rule.

62. The method of claim 57, wherein the first party is protected by the order, further comprising recording, on the device only, detections of the proximity tokens of the second party following suggestions, for submission at the first party's election.

63. The method of claim 57, further comprising receiving a timetable assigning the first party time windows for a place used by both parties, without disclosure of the time windows of the second party.

64. The method of claim 59, further comprising importing to the device, with the first party's consent, a class schedule, work schedule, or calendar of the first party; expanding recurring events into planned space-time cells for a period of days or weeks; classifying an intersection as recurring, one-off, or in transit; and, for a recurring intersection, sending through an intermediary to an administrator a request for an accommodation, such as a change of section, shift, entrance, or attendance mode, that discloses neither the identity nor the schedule of the second party.

65. The method of claim 57, further comprising periodically receiving from the first party a private choice to keep, increase, or reduce the degree of separation, wherein an increase chosen by either party takes effect without disclosing which party chose it, and a reduction takes effect only if both parties chose a reduction, neither party otherwise learning the other's choice.

66. A method performed by one or more computing devices, the method comprising: maintaining, for parties to a no-contact order, a degree of separation enforced by suggestions presented on the parties' own devices; at review intervals, receiving from each party a sealed choice to keep, increase, or reduce the degree of separation; applying an increase chosen by either party without disclosing which party chose it; applying a reduction, in graduated steps each revertible by either party, only where both parties chose a reduction; and, where both parties did not choose a reduction, disclosing to neither party the other party's choice.

67. The method of claim 57, wherein a degree of separation is selected from an ordered set of levels of increasing restriction, comprising at least proximity suggestions alone, proximity suggestions with protected zones, and schedule-based avoidance, starting at the least restrictive level that meets a need stated by the protected party, increasing only upon a request and a finding of necessity, and proposing a lower level after a configured period without conflicts.

68. The method of claim 57, further comprising determining, before the proximity tokens are recognized, that the parties are within a satellite-positioning-scale distance of each other by private proximity testing, in which each device derives, from its own position, coarse location cells salted

with the pair secret and a current time window, and the devices learn only whether any cell is shared, neither device learning the other's position; and presenting a route or timing suggestion upon such a determination and a closer-range suggestion upon recognition of a proximity token.

69. A method performed by one or more computing devices including a user device having a microphone, the method comprising: detecting that an arming cue has been given, the arming cue being an ordinary input that does not name a safety function, designated in advance by a user or by a default configuration, and selected from a media selection, a request to a voice assistant, a command to a connected device, a button or touch pattern, a gesture, and a placement of a device; responsive to the detecting, carrying out any ordinary response to the arming cue and entering a guard state; in the guard state, detecting, without first detecting a wake word, a safe phrase in audio captured by the microphone; and, upon detecting the safe phrase, presenting a private check-in on a device of the user and, upon absence of a response to the private check-in, initiating a help response.

70. The method of claim 69, wherein the safe phrase comprises a universal emergency phrase adopted under a policy applicable to a locale of the user device, detected regardless of whether the user has enrolled any safe phrase of the user's own.

71. A method performed by a mobile computing device of a first user of a platform, the method comprising: upon the first user blocking a second user on the platform, or the second user blocking the first user, establishing with a device of the second user a pair secret not known to any server of the platform; broadcasting proximity tokens derived from the pair secret and a current time, the tokens being unlinkable by devices that do not hold the pair secret; recognizing a proximity token broadcast by the device of the second user, or determining by private proximity testing over location cells salted with the pair secret that the device of the second user is within a configured distance or that a venue planned by either user is shared; and, responsive thereto, presenting on the mobile computing device a suggestion to change route, venue, or timing without indicating the location, identity, or presence of the second user; wherein no location of either user is transmitted to a server of the platform or to the other user.

72. A method performed by a first device of a plurality of devices bound to a user, the method comprising: entering a guard state upon an arming input of the user, or, where the user has so elected, upon detecting by short-range wireless detection that a device of a second person, other than a device of the user, is present in a bedroom of the user, the guard state being a state in which no affirmation is awaited and no output is conditioned on any affirmation; transmitting, over a local

link and authenticated by a key shared among the plurality of devices, an arming message causing each other device of the plurality present in the bedroom to enter the guard state without any change perceptible to the second person; in the guard state, detecting on the first device, without first detecting a wake word, a safe phrase of the user, or receiving from another device of the plurality an indication that it detected the safe phrase; and, responsive thereto, coordinating among the plurality of devices an escalation comprising stopping any playback, a private check-in on a device of the user, an alarm, and a call for help through a pathway pre-configured by the user, each stage being rendered by a device of the plurality selected according to its capabilities, and the escalation advancing only upon absence of a response from the user; the guard state ending on the plurality of devices only upon closing by the user.

73. A mobile computing device comprising a processor, a microphone, a wireless interface, a positioning subsystem, and memory storing instructions that, when executed by the processor, cause the device to perform the method of any one of claims 44 to 46, 57 to 65, 67 to 72, 77 to 85, 86 to 95, and 97 to 108.

74. A smart speaker comprising a far-field microphone array, a loudspeaker, a wireless interface, a processor, and memory storing instructions that, when executed by the processor, cause the smart speaker to perform the method of any one of claims 69, 70, 72, 77 to 80, 86 to 95, 97 to 104, and 106 to 108.

75. A system comprising one or more processors and memory storing instructions that, when executed, cause the one or more processors to perform the method of any one of claims 47 to 56 and 66.

76. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause the one or more processors to perform the method of any one of claims 44 to 72, 77 to 85, 86 to 95, and 97 to 108.

77. The method of claim 69, wherein the guard state is entered only while the user device is determined to be in a bedroom of the user from a designation learned on the user device by: detecting, over a plurality of sleep periods of the user, each sleep period from at least one of a charging cycle of a phone or wearable device of the user during night hours, a sleep or wake cycle of the user, a wake-up alarm set at night and turned off after an interval consistent with sleep, a sleep signal sensed by a wearable device of the user, and recurring snoring of which only a count or flag is retained; recording during each sleep period a fingerprint of the place where the user device or a phone of the user rests, comprising at least one of signal strengths of nearby wireless networks,

a barometric pressure, and a local magnetic field; and designating the place as the bedroom upon recurrence of the fingerprint over a configured number of sleep periods and confirmation by the user.

78. The method of claim 69, wherein the guard state is entered only while the user device is determined to be in a bedroom of the user from a label, entered by the user in an application, designating a place, a room, or a fixed device as the bedroom of the user.

79. The method of claim 69, wherein the guard state is entered only while the user device is determined to be in a bedroom of the user from a designation learned on the user device from signals obtained without a location permission, a Bluetooth permission, a health-data permission, or a microphone permission, the signals comprising at least two of a time of night, a charging state of a phone of the user, a motion and orientation of the phone reported by its accelerometer or gyroscope, an ambient light level, and a barometric pressure.

80. A method performed by a user device, the method comprising: computing, on the user device, a probability that the user device is in a bedroom of a user by combining a plurality of signals, each weighted according to how strongly it indicates a bedroom, the signals comprising at least two of a charging state of a phone of the user during night hours, a motion and orientation of the phone, an ambient light level, an ambient sound level from which no audio is retained, a barometric pressure, a fingerprint of nearby wireless signals, a wake-up alarm set at night, a sleep signal sensed by a wearable device, a name or group of a room or device in a smart-home application, a night-time routine of a smart-home system, playback of sleep sounds at night, occupancy reported by a bed sensor, presence reported by a non-imaging sensor, an acoustic response of the room measured by a loudspeaker and a microphone of a fixed device, a pattern of use of a fixed device comprising at least one of alarms, sleep timers, and night-time or morning routines, an absence of speech at a fixed device through night hours, rooms or names assigned to other fixed devices of the household, and an ambient light level at a fixed device; upon the probability exceeding a threshold, designating the place as the bedroom of the user upon confirmation by the user or, where the user has so elected, automatically; and, while the user device is determined to be in the designated bedroom, detecting on a device of the user, without first detecting a wake word, a safe phrase in audio from a microphone, and, upon detecting the safe phrase, executing a response comprising at least one of a check-in, an alarm, and a call for help through a pathway pre-configured by the user.

81. The method of claim 80, further comprising: determining that the user and a second participant are in an early stage of a relationship from at least one of an indication by each of them, a match

between them on a linked dating platform within a configured period, and an absence of a prior designation of each other as established partners; while the early stage is determined, opening an affirmative consent envelope in which playback of media continues only while an affirmation from each participant is received within an affirmation window, and, upon a lapse of the affirmation window, fading the playback and issuing a check-in; and, upon each participant privately electing on that participant's own device that the envelope no longer be applied between them, operating instead in a guard state in which no affirmation is awaited and no playback is gated, the safe phrase and the response remaining armed, and any participant being able to restore the envelope at any time.

82. The method of claim 81, further comprising: determining, from at least one of a stage of the relationship, a time since the participants matched or designated each other as established partners, a stated preference of a participant, and a recent event, a recommended setting of the affirmative consent envelope selected from keeping the envelope, moving a phaseout point of the envelope defined by at least one of an elapsed time, a stage of a session, and a number of renewals, and enlarging the affirmation window to span an entire session; presenting the recommended setting privately to each participant, on that participant's own device, for affirmation, at a time selected from the start of a session, during a session, and a later check-in after a session; and applying a recommended setting that requests fewer renewals only when every participant has affirmed it, the most protective setting applying where the responses differ, and applying a setting that requests more renewals upon the affirmation of any one participant.

83. The method of claim 80, further comprising, in a solo mode in which only the user is enrolled: opening an affirmative consent envelope in which playback of media continues only while an affirmation of the user is received within an affirmation window; determining a recommended setting of the envelope selected from keeping the envelope, moving a phaseout point of the envelope, and enlarging the affirmation window to span an entire session; presenting the recommended setting to the user for affirmation at a time selected from the start of a session, during a session, and a later check-in after a session, the setting taking effect upon the user's affirmation; and honoring a stop word spoken by any person present, including a person who is not enrolled; wherein the safe phrase and the response remain armed under every setting.

84. The method of claim 82, wherein the recommended setting is presented with the setting preselected and is accepted by a single input selected from a tap, a button press, a touch on a wearable device, and a short spoken word, the presenting further offering an option to edit the

recommended setting before acceptance; wherein absence of a response leaves the current setting unchanged; and wherein, for a participant who has elected automatic acceptance, a recommended setting that keeps or increases the renewals requested is applied with a private notice and without awaiting that participant's input.

85. The method of claim 83, wherein the recommended setting is presented with the setting preselected and is accepted by a single input selected from a tap, a button press, a touch on a wearable device, and a short spoken word, the presenting further offering an option to edit the recommended setting before acceptance; and wherein, where the user has elected automatic acceptance, the recommended setting is applied with a private notice unless the user declines it.

86. The method of claim 69, wherein the guard state is entered only while the user device is determined to be in a bedroom of the user from at least one of an assignment of the user device to a bedroom in a smart-home application, a room tag of the bedroom read by the user device, and a designation learned from where the user device rests during sleep periods of the user.

87. The method of claim 69, wherein the safe phrase is detected on the user device in audio from which any media being played by the user device has been cancelled using that media as a reference.

88. The method of claim 69, wherein entering the guard state produces no output other than the ordinary response and a private acknowledgement on a device of the user.

89. The method of claim 69, wherein, in the guard state, no affirmation is awaited and no playback is conditioned on any affirmation.

90. The method of claim 69, further comprising, upon detecting the safe phrase, stopping any playback by the user device.

91. The method of claim 69, wherein the help response comprises sounding an alarm upon the absence of a response to the private check-in and placing a call for help through a pathway pre-configured by the user upon absence of a response to the alarm.

92. The method of claim 69, wherein the private check-in is delivered on a plurality of devices of the user including a wearable device, is repeated at least once with increasing intensity before the absence of a response is determined, and is answerable by a single input selected from a tap, a squeeze, and a short phrase in a verified voice of the user, and wherein a duress answer that appears identical to an all-clear answer initiates the help response.

93. The method of claim 69, wherein the help response first notifies a contact designated in advance by the user and requests emergency services only upon at least one of an answer of the user requesting help, a duress answer, a corroborating distress signal selected from a scream, an impact, a pattern consistent with freezing, and a physiological distress signal of the user, and a failure to reach any designated contact within a configured period.

94. The method of claim 93, further comprising attaching to a request to emergency services a validation level computed from the evidence detected, the validation level distinguishing at least a safe phrase followed only by an absence of a response from a safe phrase accompanied by a corroborating distress signal or by an answer requesting help.

95. The method of claim 69, wherein a detection of the safe phrase is accepted only when its confidence exceeds a threshold and at least one of the safe phrase being repeated within a configured interval and a voice of the safe phrase matching an enrolled voice of the user, and wherein repeated detections that the user dismisses as mistaken cause the user device to prompt the user to choose a different safe phrase, without lowering protection after any detection that the user did not dismiss.

96. A system comprising a user device having a microphone and one or more further computing devices selected from a phone, a wearable device, a smart speaker, and a server, the user device and the further computing devices together comprising processors and memory storing instructions that, when executed, cause the system to perform the method of any one of claims 69, 70, 77 to 79, 86 to 95, and 97 to 108, each step of the method being performed by any one or more of the devices of the system.

97. The method of claim 69, wherein the arming cue is a media selection and the ordinary response comprises playing the media selection, the method further comprising, while the media selection plays, preventing speech and vocals within the media selection from being detected as the safe phrase, and, upon detecting the safe phrase and where the user has so elected, fading the media selection to a reduced level, the fade serving as an ambient signal perceptible in the room that a check-in is pending, and resuming the media selection upon an all-clear answer to the private check-in.

98. The method of claim 69, wherein the arming cue is a media selection and the ordinary response comprises playing the media selection, and wherein, where the user and each other participant have so elected, entering the guard state further comprises opening an affirmative consent envelope in which the media selection continues to play only while an affirmation from each participant is

received within an affirmation window, a lapse of the affirmation window fading the media selection and issuing a check-in.

99. The method of claim 98, wherein a volume of the media selection is set by a function, linear or nonlinear, of a current consent score of the participants, the volume following the lowest current consent score among the participants, and a consent score below a threshold fading the media selection out and issuing a check-in, whereby the volume serves as an ambient indication of the current consent score.

100. The method of claim 99, wherein the consent score is computed from tiers of enrolled go words of the participants, a higher tier producing a higher volume within a range set by the participants.

101. The method of claim 99, wherein, absent a deliberate affirmative signal of a participant received through any input enrolled by that participant, the current consent score of that participant decays gradually over time, lowering the volume as an ambient prompt for renewal, and a renewed affirmative signal restores the consent score, and wherein an involuntary physiological signal does not raise the consent score.

102. The method of claim 69, wherein the arming cue is a media selection played by a device or service other than the user device, the method further comprising recognizing the media selection, on the user device, by matching an acoustic fingerprint of audio captured by the microphone against a fingerprint of the media selection enrolled as the arming cue, and, in the guard state, identifying the track and position being played, obtaining corresponding audio of the track from a catalog or a music service of the user, aligning it with the captured audio, and cancelling it from the captured audio using the aligned audio as a reference before detecting the safe phrase.

103. The method of claim 70, further comprising receiving at the user device, over a short-range wireless broadcast from a nearby device not bound to the user, an anonymous alert that the nearby device detected the universal emergency phrase, the alert carrying no identifier of any person or account, and treating the alert, while the guard state is active, as a detection of the safe phrase, whereby the private check-in and the help response are initiated by the user device.

104. A method performed by a device having a microphone, the method comprising: detecting on the device, without first detecting a wake word, a universal emergency phrase adopted under a policy applicable to a locale of the device; and, responsive thereto, broadcasting over a short-range wireless link an alert that the universal emergency phrase was detected, the alert carrying no

identifier of any person or account and being broadcast under a rotating identifier of the device, in a format by which a nearby device of another manufacturer, whose user has a guard state active, treats the alert as a detection of a safe phrase and initiates a private check-in on a device of that user.

105. A method performed by a mobile device, the method comprising: receiving at the mobile device, over a short-range wireless broadcast from a nearby device not bound to a user of the mobile device, an anonymous alert that the nearby device detected a universal emergency phrase, the alert carrying no identifier of any person or account; responsive to the receiving, presenting on the mobile device, including where no guard state has been armed on the mobile device, a private check-in comprising an option to request help; upon selection of the option, initiating a help response comprising at least one of notifying an emergency contact and initiating an emergency call, each as configured in an operating system of the mobile device; and, upon absence of a response to the private check-in, initiating no help response.

106. The method of claim 69, wherein, while media plays within range of the microphone, a candidate detection of the safe phrase is rejected where it is classified, from acoustic features of audio captured by the microphone, as reproduced by a loudspeaker rather than spoken live, the acoustic features comprising at least one of a frequency response or band limit of a loudspeaker, dynamic-range compression, steady or pitch-corrected pitch, a steady beat or accompanying instruments, a direction of arrival matching a location of a device playing the media, and an absence of near-field effects of a voice close to the microphone, and wherein a candidate detection whose classification is uncertain is accepted.

107. The method of claim 69, further comprising detecting, in audio captured by the microphone, a watermark embedded in media playing within range of the microphone, authenticating the watermark on the user device, and rejecting a candidate detection of the safe phrase only where the watermark is authenticated and words of the candidate detection match lyrics or a transcript of the media at a playback position indicated by the watermark, whereby an unauthenticated watermark does not suppress detection of the safe phrase and listening continues for the duration of the media.

108. The method of claim 69, wherein the arming cue is a media selection played by a device or service other than the user device, the method further comprising recognizing the media selection, on the user device, from at least one of a watermark embedded in the media selection and detected in audio captured by the microphone, and metadata identifying the media being played that is received from the playing device or from a casting, shared-listening, or media-control session to which the user device is connected.

ABSTRACT OF THE DISCLOSURE

Mechanisms reduce risk around intimate encounters. A device in a user's bedroom, recognized from a label or sleep-period signals, is armed discreetly by an ordinary cue, such as a song or a routine voice request, and enters a guard state: it listens on the device, without a wake word, for a safe phrase while cancelling its own playback, and on detection runs a private check-in, an alarm, and a call for help, each advancing only without a response. A phone arriving late at night at an unfamiliar residence in a risk context presents a private graded check-in. A refuge service screens a discreet request for medical risk and assigns a safe-sleep room or vetted host with a time-limited credential. An intermediary carries screened, sealed apologies under a no-contact order with verifiable transmittal records, and the parties' devices suggest routes that keep them apart without sharing locations.